

Survey Paper

An overview of Eulerian video motion magnification methods[☆]Ahmed Mohamed Ahmed^{a,*}, Mohamed Abdelrazek^b, Sunil Aryal^a, Thanh Thi Nguyen^a^a School of Information Technology, Deakin University, Victoria, Australia^b Applied Artificial Intelligence Institute, Deakin University, Victoria, Australia

ARTICLE INFO

Keywords:

Motion magnification

Eulerian

Lagrangian

Deep convolutional neural network

ABSTRACT

The concept of video motion magnification has become increasingly relevant due to its ability to detect small and invisible motions that can be of great value in a variety of applications. A variety of approaches have been developed to magnify these motions and variations. While both Eulerian and Lagrangian processing methods are widely used for motion magnification, Eulerian approaches are more commonly employed due to their lower computational cost. This paper provides an overview of the powerful Eulerian motion magnification techniques. We begin with a brief introduction to technical concepts associated with Eulerian motion techniques such as pyramids and filters in image processing. Additionally, we provide a comparison between the Lagrangian and Eulerian perspectives, followed by a comprehensive overview of the various Eulerian motion magnification (EVM) techniques available. Finally, we present implementation results and a comparative analysis of some of the Eulerian motion techniques.

1. Introduction

In the present-day world, there exist many subtle motions and color changes that remain undetected by the human eye but could be highly beneficial for diverse applications in various fields, including but not limited to medical monitoring, clinical diagnosis, physiological functions, construction monitoring, forensic analysis, and security. To this end, motion magnification techniques have been developed to uncover and amplify these subtle variations in videos that are otherwise invisible to the naked eye. For instance, in the context of medical monitoring and clinical diagnosis, the human skin undergoes color changes owing to blood circulation, and video magnification techniques such as those proposed by [1] can be used to monitor the pulse rate of patients or elderly people. Similarly, motion magnification techniques [1,2] have been applied to detect and amplify the small breathing motions of infants, making them clearly visible. In a similar vein, small motions in critical machines can provide valuable insights into their mechanical behaviors and help monitor the quality of mechanical systems.

Motion magnification techniques, such as those presented in works by Wu et al. [1], Wadhwa et al. [2,3], and Oh et al. [4], have wide-ranging applications across various fields. They are instrumental in detecting subtle structural movements and facilitating structural health monitoring in scenarios like swaying buildings and bridges, as demonstrated in studies by [2,5–8]. Moreover, these techniques find applications in the biomedical domain [1,9–15]. Beyond these

applications, video motion magnification is also valuable in fields like security, where it aids in recognizing facial movements and micro-expressions [16–18], contributes to deepfake detection [19–22], and serves various other domains [23–26].

In essence, motion magnification techniques function as a virtual microscope that enables the visualization of small and imperceptible motions in videos. Unlike conventional microscopes that rely on optics, motion magnification techniques are software-based programs that take an input video and output a magnified version of the temporal variations present within it. The amplified output generated by these techniques can be exploited in various applications without the need for additional equipment, which can incur higher operating expenses and may require regular maintenance. In this sense, motion magnification techniques offer a cost-effective and low-maintenance alternative to traditional magnification approaches, making them a highly attractive option for researchers and practitioners across different domains.

Recent motion magnification techniques can be classified into Lagrangian and Eulerian perspectives. Among Eulerian methods, the Eulerian video magnification (EVM) technique can be categorized as either linear or phase-based. The linear EVM approach uses a first-order expansion of the Taylor series, which simplifies motion detection in videos and allows for fast detection of small motion variations. However, this method can fail when the magnification factor is large, and the Taylor approximation becomes inaccurate. In contrast, phase-based magnification (PVM) uses a Fourier decomposition by a complex

[☆] This article was recommended for publication by Prof. A Serrano.

* Corresponding author.

E-mail addresses: ammohamed@deakin.edu.au (A.M. Ahmed), mohamed.abdelrazek@deakin.edu.au (M. Abdelrazek), sunil.aryal@deakin.edu.au (S. Aryal), thanh.nguyen@deakin.edu.au (T.T. Nguyen).

Table 1
Summary table comparing various motion magnification methods. Sourced from [4,31].

Method	Liu et al. [32]	Wu et al. [1]	Wadhwa et al. [2]	Wadhwa et al. [3]	Zhang et al. [30]	LB-VMM [4]	STB-VMM [31]
Spatial decomposition	Tracking, optical flow	Laplacian pyramid	Steerable filters	Riesz pyramid	Steerable filters	Deep convolution layers	Swin Trans
Motion isolation	–	Temporal bandpass filter	Temporal bandpass filter	Temporal bandpass filter	Temporal bandpass filter (2nd-order derivative)	Subtraction or temporal bandpass filter	Subtraction
Representation denoising	Expectation maximization	–	Amplitude weighted Gaussian filtering	Amplitude weighted Gaussian filtering	Amplitude weighted Gaussian filtering	Trainable convolution	Swin Transformer

steerable pyramid [27,28] to replace the linear approximation. Although PVM is more complex and time-consuming, it can support larger motion amplification. Both linear and phase-based EVM approaches are faster and less affected by noise compared to Lagrangian-based video magnification. However, EVM approaches are not suitable for large and arbitrary motions [29].

Numerous studies have delved into motion magnification, resulting in the development of various techniques tailored to specific applications. This paper primarily focuses on Eulerian video magnification (EVM) methods, initially introduced by Wu et al. [1] and subsequently refined in works such as [2–4,30,31], among others. Although the primary aim of this paper is not to delve into the extensive array of applications across diverse fields where these methods have found utility, we will briefly outline some application domains in the following sections to illustrate the effectiveness and versatility of EVM techniques. To provide a comprehensive understanding of this method, we offer a brief overview of one of the Lagrangian approaches [32] to highlight the key differences between Eulerian and Lagrangian perspectives and to justify our preference for the former. Moreover, in the background section, we elucidate various image processing technologies such as pyramids and temporal filters, which are intelligently incorporated in motion magnification techniques, and explain the underlying principles of Lagrangian and Eulerian perspectives. We then provide an analysis of the primary approaches in Eulerian motion magnification, along with their respective advantages and drawbacks. Finally, we present implementation results and an empirical comparison between the methods to demonstrate the efficacy of EVM in various applications (see Table 1).

2. Related works

Motion magnification techniques have been classified into Lagrangian and Eulerian methods. Lagrangian methods are based on tracking motion, which involves following specific points or regions of interest in the video sequence over time. This process can be time-consuming and computationally expensive, especially for long videos or high-resolution data [33–35]. On the other hand, Eulerian methods are based on measuring the flow of motion at a fixed location, which can be considered a field-based method. Both Lagrangian and Eulerian methods have exhibited the potential to extract subtle motion signals from video data with minimal manual intervention. However, our primary focus in this paper centers on the Eulerian perspective. Specifically, we delve into the most prominent and widely-utilized Eulerian video magnification (EVM) approaches [1–4,30,31]. While we will not explore the extensive range of applications and diverse fields where these methods have been implemented, we will briefly touch upon some applications and use cases in this section and in Section 5. These examples serve to illustrate the effectiveness and efficiency of EVM techniques and their practical and potential applications across various domains.

One prominent contribution to the field of Eulerian video magnification is the method developed by Wu et al. [1]. This method

involves decomposing the input video into various spatial frequency bands and then extracting the frequency bands of interest, which are subsequently amplified by a specified factor. This pioneering work paved the way for further research in this area, including the use of linear Eulerian video magnification in a wide range of applications, including healthcare monitoring, animal physiology, and several other applications [11,36–38].

The initial Eulerian video magnification approach presented by Wu et al. [1] was simple and efficient. However, this method has its limitations, particularly when it comes to amplifying large input motion or high magnification factors. To address these limitations, Wadhwa et al. [2] introduced the first phase-based Eulerian approach, which proved to be more effective than the linear approach. In a subsequent study, Wadhwa et al. [3] presented a phase-based magnification approach that utilized Riesz pyramid, which allowed for real-time processing and made the method faster. While Wadhwa et al.'s work [2,3] primarily concentrates on visualizing subtle motions, Davis et al. [39] emphasize measuring these motions and utilizing them for sound recovery. The local motion signals in their study are obtained from phase variations within the complex steerable pyramid introduced by Simoncelli et al. [27,28], as these variations have proven effective in capturing minor motions in video data [2].

Zhang et al. [30] followed in the footsteps of Wu et al. [1] and Wadhwa et al. [2], but they introduced acceleration magnification to address the problem of large motion. The approach presented in the study by Takeda et al. [40] is inspired by the work of Zhang et al. [30] but employs the use of jerk as a fundamental technique. In [41], the authors employ a fast local Laplacian filter [42], while Wu et al. [43] incorporated Principal Component Analysis (PCA) into video magnification. In their study [44], the researchers employ Fractional Anisotropy (FA) to create a unique filter, termed the fractional anisotropic filter, which selectively amplifies meaningful subtle changes.

These phase-based Eulerian methods have shown great potential in a variety of applications. For instance, they have been used for recognizing micro-expressions, human activities, biomedical and various other applications [13,45–52]. Takeda et al. [53] introduce a bilateral video magnification filter (BVMF) that operates in both the temporal and intensity domains, similar to a bilateral filter [54–56]. One kernel conducts temporal bandpass filtering using a Laplacian of Gaussian, focusing on the target frequency with unity gain. The other kernel employs Gaussian filtering on the signal intensity through the Fourier shift theorem to eliminate large motions beyond the specified magnitude. However, their approach still depends on hand-designed filters that may potentially not effectively address swift, large motions.

The conventional approach for Eulerian video magnification is to decompose the input video sequence into representations using hand-designed filters. However, Oh et al. [4] introduced a novel method that uses deep convolutional neural networks to learn the decomposition filters. This approach has the potential to improve the accuracy and effectiveness of Eulerian video magnification by leveraging the powerful learning capabilities of deep neural networks. The learned filters

can adapt to the specific features and characteristics of the input video, potentially leading to accurate magnification results.

Numerous subsequent studies have drawn inspiration from Oh et al. [4], integrating motion magnification techniques with deep learning. For instance, in their recent work [31], the authors present Swin Transformer Based Video Motion Magnification (STB-VMM), building upon the Learning-based motion magnification VMM (LB-VMM) [4] but incorporating the Swin Transformer architecture [57] to reduce artifacts. Additionally, Fei et al. [19] propose a method centered on motion discrepancies to effectively distinguish AI-generated fake videos from authentic ones, employing a two-stage architecture consisting of a CNN+LSTM.

Das et al. [20] integrate traditional Eulerian video magnification [1] with deep learning models in their research. They utilize features extracted from the Eulerian technique to train three models for classifying altered and unaltered videos. Additionally, Ciftci et al. [21] propose an approach that capitalizes on the discrepancy between genuine motion and amplified GAN fingerprints by combining deep learning [4] with traditional motion magnification [2]. In another study, Albor et al. [58] merge traditional methods with the Hermite Transform and deep learning to monitor heart and respiratory rates. Moreover, Lado Roig et al. [6] employ learning-based video motion magnification [4] to automatically extract vibrational signatures, aiding in the identification of structural damage. Additionally, papers by Singh et al. [59,60] utilize deep learning-based solutions for motion magnification, with the goal of mitigating artifacts.

Numerous publications have applied Eulerian video magnification (EVM) methods across diverse application domains. While these studies may not introduce new techniques, they serve the essential purpose of demonstrating the practicality and versatility of EVM in various fields. Although our focus in this paper is primarily on novel concepts and enhancements to existing methods such as [1–4], we find it valuable to provide select examples of applications from the extensive array of fields where these techniques have proven useful.

Motion magnification techniques have found practical application in contactless structural monitoring and structural health assessment in recent studies [5–7,49,52,61]. These vision-based monitoring methods, particularly motion magnification, offer the advantage of eliminating the need for physical installations and contact, which is especially beneficial for structures that are challenging to access or pose safety concerns [8]. For instance, in [49,50], the authors drew inspiration from the Phase-Based motion magnification technique [2] to develop an approach for structural health monitoring (SHM) systems applicable to both existing and new structures. In [51], researchers explored the estimation of material properties using video data. Additionally, motion magnification has been employed in [5] for monitoring ancient constructions, while in [6], it was utilized for damage detection in structures. Furthermore, [7] applied motion magnification to civil engineering structures, and in [52], the authors employed phase-based video magnification for damage detection and structural health monitoring.

These motion magnification (MM) techniques have found applications in the field of biomedical and healthcare [9–15]. In a particular study [36], researchers applied color magnification to amplify subtle color variations caused by blood pulse and compared these results with those obtained from pulse sensors. In another work [38], Eulerian Motion Magnification combined with principal component analysis (PCA) was employed to extract breathing waveforms from upper body movements, enabling the rapid and precise extraction of various breathing patterns from video data. Moreover, the study by [47] introduces the use of Phase-Based Eulerian Video Magnification (PB-EVM) to detect small breathing movements in human bodies.

Additionally, the study [58] introduced a method that combines traditional techniques with the Hermite Transform and deep learning for monitoring heart and respiratory rates. Henrik et al. [11] applied Eulerian Video Magnification (EVM) [1] to reveal concealed physiological

information, including heart and breathing rates, in animal videos. This was done to demonstrate that vocalization-induced vibrations could be revealed in Asian infrasonic elephants. The work [62] used MM for Respiratory Rate Estimation. Furthermore, EVM has been used in visual microscopy of biological cells, as shown by Shabi et al. [12], to amplify sub-micron rhythmic motions that are not visible with conventional microscope lenses, such as fluorescent microscopy technology [63]. Meanwhile, in the work by [64], motion magnification was utilized for video-based breathing monitoring, and in the work [65], they applied MM [2,66] to detect spontaneous tympanic membrane displacement as an indirect measure of intracranial pressure in patients with brain pathologies.

Eulerian video magnification has also proven valuable in security applications, particularly in the domain of facial micro-expression recognition [16]. In Peng et al.'s study [17], the MM method was employed to reveal subtle facial movements and expressions. Similarly, in Bai et al.'s work [18], VMM methods were utilized to amplify minor facial movements, aiding in the micro-expression recognition task, and also contribute to deepfake detection [19–22]. Motion magnification has demonstrated its effectiveness in a wide range of applications in various fields, as evidenced in studies and works such as [23,24], and the study focused on infrastructures [8].

While our paper primarily focuses on Eulerian approaches, it is worth noting that there exist intriguing motion magnification methods and applications that diverge from the Eulerian framework. These approaches may adopt a Lagrangian perspective or combine elements of both Lagrangian and Eulerian techniques. Lagrangian video magnification also has witnessed significant advancements through several notable approaches, which have found applications in various fields [32,67–71]. Liu et al. [32] proposed a mechanism that tracks particle trajectories over time and analyzes the trajectories of feature points. In [67], they developed a filtering technique that modulates input motion to enhance the animation and liveliness of the output motion. In the domain of facial micro-expression recognition, the authors in [68] employed the Lagrangian perspective, demonstrating improved recognition performance compared to local Eulerian-based approaches, particularly when dealing with higher magnification factors. This is particularly noteworthy as it addresses one of the limitations associated with Eulerian video magnification techniques. The contributions of these Lagrangian approaches have expanded the application potential of video magnification and opened avenues for further research and development in this field.

In Flotho et al. [69], a Lagrangian motion magnification technique is employed, relying on dense optical flow estimation for psychophysiological experiments. In another work by Flotho et al. [70], they propose a local Lagrangian motion magnification method specifically designed for enhancing facial micromovements. This approach fine-tunes the recurrent all-pairs field transforms for optical flow (RAFT) deep learning method [72]. Also, in the study by Balakrishnan et al. [9], heart rate and beat lengths are extracted from videos by monitoring subtle head motion resulting from the Newtonian response to blood influx during each heartbeat. This method involves tracking head features, applying principal component analysis (PCA) to decompose their trajectories into distinct component motions, and selecting the component most aligned with heartbeats based on its temporal frequency spectrum. Subsequently, motion projected onto this component is analyzed, and peaks in the trajectories are identified as heartbeats. Evaluation involving 18 subjects revealed heart rate measurements nearly identical to those obtained with an electrocardiogram device.

There are advanced methods that utilize intricate motion-tracking algorithms to tackle specific challenges. One such approach is presented by Elgharib et al. [73], who have combined both Lagrangian and Eulerian techniques to address the issue of quality degradation caused by significant motion in the input video sequence. The proposed method, called Dynamic Video Motion Magnification (DVMAG), consists of two layers. The first layer involves image warping, which removes large

motion while avoiding any artifacts, such as by utilizing the KLT tracking algorithm [74]. The second layer applies layer-based magnification. Although DVMAg has demonstrated high-quality results, it requires significant complexity in terms of implementation and practical use.

In their research [75], various methods were employed to identify objects within video sequences. These techniques included kernel K-means for segmentation, automatic scribble drawing using superpixels and Bezier curves, alpha matting, and Eulerian motion magnification. However, it should be noted that this approach necessitates user intervention to select the region where subtle changes should be amplified. The work by Gao et al. [76] introduces a hybrid video motion magnification method named MagFormer, combining elements of both Eulerian and Lagrangian approaches. MagFormer employs an end-to-end architecture with a two-branch module that integrates Convolutional Neural Networks (CNN) and Transformer [77,78]. The Eulerian branch focuses on extracting local information, while the Lagrangian branch captures global information. Additionally, they propose the use of a motion-guided mask to select specific motion components in videos. However, it is a complex and computationally expensive method due to its utilization of the Lagrangian perspective. In [79,80], the authors employ multi-task learning to combine Lagrangian and Eulerian methods. Their approach primarily builds on Eulerian methods for efficient inference while incorporating optical flow from Lagrangian methods to enhance precise motion perception.

This paper predominantly concentrates on motion magnification methods within the Eulerian perspective, excluding the Lagrangian perspective [67–70] and other non-Eulerian techniques [81]. However, to provide a comprehensive overview, we briefly discuss one Lagrangian approach [32] as an illustrative example, highlighting its higher computational complexity and cost compared to the Eulerian approaches. Furthermore, in this paper, we do not cover older approaches that utilized similar techniques to motion magnification for manipulating and re-displaying motion data. Examples include modifying motion to create new activities [82,83] or altering the style of videos [84]. Some projects focused on video motion analysis for output synthesis, such as imitating motion blur [85], identifying temporal jumps in video textures [86], or employing layered motion analysis for video synthesis [87,88]. While these approaches have their own objectives and relevance, they are not the primary focus of our paper.

3. Background

The background section of this paper provides an overview of the concepts and techniques that are commonly used in video processing and motion magnification. The section will cover the Taylor series [89–91], which is a polynomial approximation method that is commonly used in linear approximation. The section will also cover pyramid representation techniques such as the Gaussian pyramid [92,93], Laplacian pyramid [92,94], wavelet decomposition [95,96], and steerable pyramid [28,97,98]. These techniques are used for multiscale analysis of images and videos and are frequently used in motion magnification algorithms to decompose the input signal into different scales or frequencies. Understanding these concepts and techniques is crucial for developing effective motion magnification methods.

3.1. A Taylor series

A Taylor series is a mathematical tool that can be utilized to represent functions as polynomials, making it a useful method in different fields of science and engineering [89–91]. The Taylor series approximation method approximates a function's values in the vicinity around a given x value. However, it is essential to note that the approach has limitations and may lead to errors in practical applications. Despite this, one of the most popular uses of the Taylor series in image and video processing is in linear approximation. Linear approximations are used to approximate functions through their tangent lines. By knowing

a function's values, say $f(d)$, and its derivative, $f'(d)$, we can determine the tangent line, which can be used to approximate $f(x)$ for points x . This approximation works well when x is relatively close to d . Therefore, the line approximation (the first-degree Taylor polynomial) for $f(x)$ for x close d will be $f(x) \approx f(d) + f'(d)(x - d)$. The Taylor series has several applications in image and video processing.

One of the areas where the Taylor series can be useful in video processing is motion magnification. Motion magnification is a technique that amplifies subtle motions in a video sequence to make them visible. The Taylor series can be used to generate an approximation of the motion field, which can be used to amplify the motions and enhance the video's details, as demonstrated in [1]. Additionally, the Taylor series can be employed for motion estimation, which is an essential task in video processing. Accurately estimating the motion of objects in a video sequence can aid in applications like video compression, video stabilization, and object tracking. By using the Taylor series, motion estimation can be performed more effectively and efficiently. So, its usefulness in linear approximation, motion magnification, and motion estimation makes it a critical tool.

3.2. Pyramids representation

In the field of image and video processing, Pyramids Representation is a commonly used technique that enables the decomposition of an image into different components, each capturing information at a specific scale. The concept of resolution and scale is closely related to the level of detail that an observer can perceive. For instance, as one approaches a distant building, they first observe an approximate outline of the building and gradually notice smaller details as they get closer. Similarly, the Pyramids Representation technique enables the analysis of images at different scales to avoid information loss at other levels [92,94].

Multi-resolution image decomposition is a crucial aspect of Pyramids Representation, which is widely used in digital image processing and computer vision [99]. It allows the analysis of objects of different sizes, each containing features of various sizes. The decomposition of images at multiple scales enables the efficient processing of information in a manner that is more accurate, and computationally efficient, and preserves the image's quality. Pyramid representation is a forerunner of scale-space representation and multi-resolution analysis, in which an image undergoes iterative smoothing and subsampling. This process results in the creation of a sequence of images, where each image has a lower resolution than the previous one, resulting in a hierarchical structure of image components that capture information at various scales. The use of the Pyramids Representation technique is advantageous in several image processing tasks, such as feature extraction, image matching, image segmentation, and image compression.

There are two primary types of pyramids: lowpass and bandpass. A lowpass pyramid is created by applying a suitable smoothing filter to the input image, followed by subsampling the smoothed image by a factor of 2 in all directions. This process is then repeated multiple times on the resulting image. Each cycle produces a smaller image with higher smoothness but lower spatial resolution. As a result, the lowpass pyramid provides a multi-scale representation of the input image, where each level captures information at a specific scale.

In Fig. 1, we can see the graphical representation of the lowpass pyramid. The authentic image is placed at the bottom, and each smaller image resulting from each cycle is placed above the input image for that cycle. This arrangement creates a pyramid-like structure, hence the name pyramid representation. On the other hand, a bandpass pyramid is created by subtracting each pyramid level from the next lower level in the pyramid. However, before the subtraction, image interpolation is required to account for the difference in sample density between the levels. The resulting bandpass pyramid contains information about the edges and details of the input image, which are not captured by the lowpass pyramid [94]. There are various types of pyramids, including the Gaussian pyramid [92,93], Laplacian pyramid [92,94], wavelet decomposition [95,96], and steerable pyramid [28,97,98], each with its specific properties and advantages.

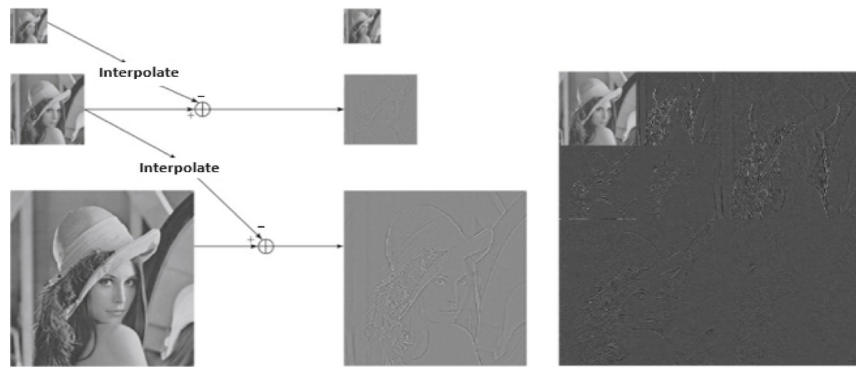


Fig. 1. Types of Pyramids: Gaussian and Laplacian (left) and Wavelet Decomposition (right). The left diagram illustrates the Gaussian pyramid, formed by repeatedly subsampling the input image to reduce resolution, and the Laplacian pyramid, created by computing differences between Gaussian filters. The right diagram shows a wavelet decomposition, which separates images into subbands, capturing details at multiple scales.

3.2.1. A Gaussian pyramid

The Gaussian pyramid is a type of multi-resolution image representation that employs a lowpass filter to decompose an image into a sequence of increasingly smaller and smoother images. The original image is convolved with a Gaussian kernel, which is a widely used smoothing filter that preserves edges while reducing noise. The resulting image is then downsampled by a factor of 2 in both the horizontal and vertical directions, resulting in an image that is half the size of the original. This process can be repeated iteratively to create a sequence of images, each with lower resolution and higher smoothness [92,93].

The Gaussian pyramid is a form of overcomplete representation, which means that the number of pixels in the pyramid exceeds the number of pixels in the original image. Specifically, if the original image has dimensions $N \times M$, then the number of pixels in the Gaussian pyramid can be approximated as $NM + NM/4 + NM/16 + \dots \approx 4/3 NM$. This is because each level of the pyramid has half the width and half the height of the previous level, resulting in a total number of pixels that is approximately $4/3$ of the original number of pixels. This overcomplete representation can be useful in some image processing tasks, as it allows for the reconstruction of the original image with higher precision than would be possible with a simple subsampling approach. Additionally, the Gaussian pyramid can be used in conjunction with other image processing techniques, such as edge detection and texture analysis, to extract information at different scales and resolutions [99].

3.2.2. A Laplacian pyramid

The Laplacian pyramid is a widely used technique in image processing that provides a multiscale representation of an image. This type of pyramid is created by subtracting each level of the Gaussian pyramid from the next lower level. In other words, the Laplacian pyramid is a bandpass filter that captures the significant details of the original image at each scale. The Laplacian pyramid is formed by decomposing the image into a series of difference-of-Gaussian images [99]. The Laplacian filter is used to enhance the edges and details of an image by amplifying the high-frequency components of the image while attenuating the low-frequency components. This makes it easier to analyze the image and detect features such as edges, corners, and textures. The Laplacian filter is applied by convolving the image with a Laplacian kernel. The result of this convolution is a Laplacian image, which highlights the edges and details of the original image [92,94].

The Laplacian pyramid is often used in image compression and coding, where it can be used to selectively remove or preserve image details at different scales. The Laplacian pyramid is also used in image blending, where it is used to combine two images into a single image by selectively merging the details of both images. Furthermore, The Laplacian pyramid is a useful tool in video motion magnification as it allows for efficient and accurate computation of the magnified motion [1].

3.2.3. A wavelet decomposition

In addition to pyramid representations, wavelet decomposition is another widely used multi-resolution technique in digital image processing [95,96]. In this method, the image is decomposed into a group of subbands, each representing details at different scales. Unlike pyramid representations, the sub-images obtained through wavelet decomposition have details with various spatial orientations. For example, edges with horizontal, vertical, and diagonal orientations are represented in different subbands. This multi-scale representation enables the detection of edges and other features at different scales and orientations, making wavelet decomposition a powerful tool in many image processing applications, such as denoising, compression, and image enhancement [99]. The wavelet transform is widely used in many practical applications, including medical imaging, image and video compression, and feature extraction for pattern recognition.

Wavelet decomposition has the ability to capture both frequency and spatial information, which makes wavelet a valuable tool in many signal and image processing tasks. Wavelet decomposition is also used in video and motion magnification. The wavelet transform can be used to analyze image or video signals in both time and frequency domains. By decomposing a video signal into different sub-bands, wavelet transform can efficiently extract the motion information at different scales and orientations. This can then be used to magnify the motion of interest while suppressing noise and other unwanted components in the signal. Therefore, wavelet decomposition can be a powerful tool in video and motion magnification [2].

3.2.4. A steerable pyramid

The steerable pyramid is a type of image decomposition technique that allows for the extraction of information in an image that is localized in both spatial scale, orientation, and position. Unlike other multi-scale and multi-orientation filters, the steerable pyramid has the advantage of being able to filter images in different directions, making it a more flexible and powerful tool for image processing. The fundamental functions used in the steerable pyramid are still defined in space and spatial frequency, similar to orthogonal wavelet transforms. However, the steerable pyramid reduces aliasing using a steerable orientation decomposition. This approach allows the pyramid to be optimized for a specific application or task, making it more adaptable than other decomposition methods [28,97,98].

Despite its advantages, the steerable pyramid has some disadvantages. For example, it is an overcomplete transform, which can lead to computational inefficiencies. However, recent advances in computer hardware and software have made it more feasible to use the steerable pyramid for a variety of image-processing tasks. The steerable pyramid is a useful tool in image processing and has been applied to a variety of applications, including image denoising, image compression, and object detection. Furthermore, the steerable pyramid is one of the

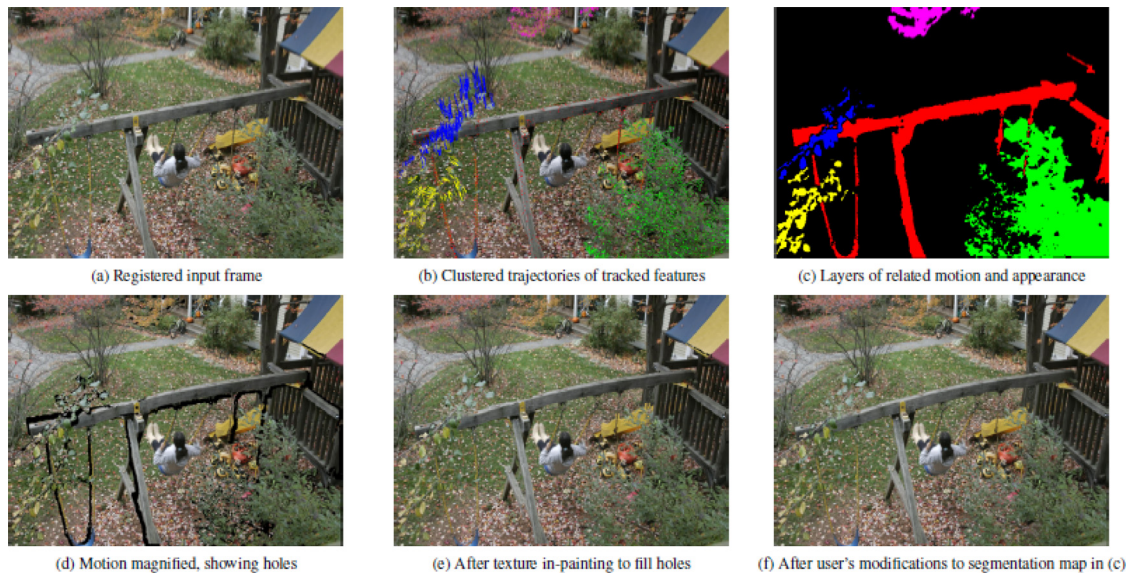


Fig. 2. Motion magnification approach steps (Lagrangian) [32].

commonly used techniques in Eulerian image and video magnification. It is used to decompose the input image or video into subbands, which capture information at various scales, orientations, and positions. These subbands can then be manipulated to amplify the motion information and reveal subtle details that are not visible to the naked eye [2,30].

4. Lagrangian and Eulerian approaches

Fluid mechanics is the study of fluid motion and Eulerian and Lagrangian approaches are two methods used to examine fluid motion. The Lagrangian approach is a method that determines a particle or a set of particles and follows them over time. This method observes fluid motion by tracking the fluid particles over space and time, just like being in a boat and drifting in a river [100,101] or following someone everywhere they go, like the character of Big Brother in the 1984 novel. However, this approach is expensive and cumbersome. On the other hand, the Eulerian approach focuses on the fluid properties at a specific location or point (x,y,z,t) . In this approach, fluid motion is observed by concentrating on certain locations, like standing on a bridge or the bank of a river and observing the flow of water at a specific and constant location, or like a manager observing an employee at a specific workplace. Therefore, Eulerian processing is suitable for working with certain technologies where the object to be tracked must be within a fixed location (such as pixel) as much as possible [100,101].

Both Eulerian and Lagrangian approaches have been used in video and motion magnification techniques such as [1–4,32]. In the case of the Eulerian approach, it can be applied to video magnification by observing the optical flow of a video sequence. Optical flow refers to the apparent motion of objects in an image or video caused by the relative motion between the observer (camera) and the objects. By analyzing the optical flow, it is possible to extract motion information and magnify it to make subtle motions more visible. Eulerian video magnification techniques can be used in medical imaging to detect subtle changes in tissue motion or in surveillance systems to detect and track small movements [1]. On the other hand, the Lagrangian approach can be used in motion magnification by tracking the motion of specific points or regions of interest over time. This approach can be applied to detect small vibrations or deformations in structures, such as bridges or buildings [32]. By tracking specific points on the structure over time, the Lagrangian approach can detect and amplify small movements to make them more visible.

In recent years, there has been an increasing interest in applying these methods to video motion magnification. In particular, the main

focus has been on the development of Eulerian video magnification approaches due to their efficiency in processing large amounts of data. These approaches involve amplifying the motion signal of a video sequence by extracting motion features at a specific location and amplifying the corresponding motion magnitudes. However, despite their advantages, Eulerian methods have limitations, such as difficulty in tracking complex and non-rigid motions, and inability to handle large displacements. In this paper, we will mainly focus on the main Eulerian Video Magnification approaches such as Linear Eulerian video magnification [1,102], Phase-Based Magnification [2,3,30], and Learning-based Motion Magnification [4], including their advantages and limitations, and present the results of the implementation of some of these methods in the following sections.

5. Main video motion magnification approaches

In this section, we will cover the key approaches in Eulerian video magnification: Linear Eulerian video magnification, Phase-Based Magnification, and Learning-based Motion Magnification. Prior to that, we will provide an overview of a Lagrangian approach [32] as an illustrative example, to emphasize the distinctions between Eulerian and Lagrangian approaches.

5.1. Motion Magnification (Lagrangian) [32]

Liu et al. [32] presented a Lagrangian perspective approach that amplifies subtle motions in video sequences. The approach acts as a microscope by analyzing and magnifying the motion, following the trajectory of particles over time, and analyzing feature point trajectories. A clustering procedure is used to automatically group several pixel clusters based on the similarity of position, intensity, and motion. A motion similarity measure is utilized to group motions based on their trajectories over time and instantaneous velocities. The approach allows the user to determine which cluster's motions should be amplified and by what factor, with small motions typically being magnified and large motions left unchanged as shown in Fig. 2. The procedure is simple yet effective in amplifying small motions and has the potential to be used in a variety of applications.

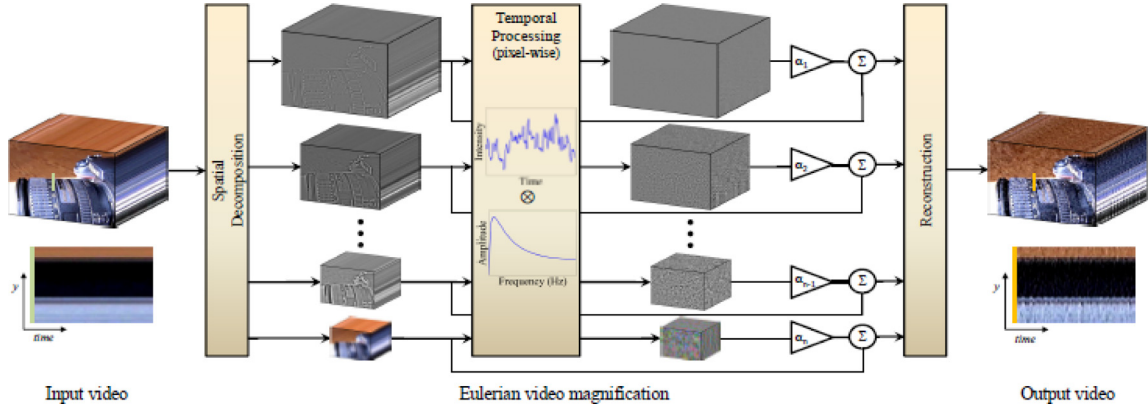


Fig. 3. Infrastructure of the linear Eulerian video magnification [1].

5.1.1. The concept

The approach involves several steps to achieve motion magnification:

Register input images: The frames are registered to correct for camera shake and eliminate small motion artifacts. Initial tracking is performed to detect the motions of a group of feature points, and outliers are discarded.

Segment feature point trajectories: Feature points are tracked over the sequence, and their paths are clustered into specific motion clusters based on similarity. This allows both small and large motions to be grouped together if they share a common cause and exhibit correlation.

Layer assignment (Segmentation): Each pixel in each frame is assigned to a suitable motion cluster or layer. Motion alone can be used for segmentation, but additional features like pixel position and closeness are considered. The Markov random by graph cuts is employed for pixel assignment, ensuring the most suitable cluster over time. The ordering of layers can be determined individually or through specific methods.

Amplify motions for chosen cluster: The user can select a motion layer for magnification from the available list of motion layers.

Render video: The rendering process starts with the background layer, which remains constant for all frames. Pixels from the outlier layer are copied as they were in the registered input. Finally, pixels from the other layers are copied to the output sequence.

5.1.2. Limitations of the approach

The presented approach has certain limitations that need to be considered. Firstly, it requires precise motion estimation, which can be challenging and may introduce errors. Secondly, the method is computationally expensive, making it less practical for real-time applications. Additionally, artifacts can occur, especially when dealing with complex motion patterns or occlusion boundaries. To achieve satisfactory results, the technique relies on additional techniques such as motion segmentation and image inpainting, which add complexity to the overall process.

5.2. Linear Eulerian video magnification

The Linear Eulerian video magnification approach, initially pioneered by Wu et al. [1], offers a straightforward and efficient method for amplifying subtle motions in videos. The technique involves several key steps as depicted in Fig. 3. First, the input video is decomposed into spatial frequency bands using an image pyramid. Then, a common temporal filter is applied to all bands to remove frame noise. Next, the filtered spatial bands are magnified by a specified factor. The resulting signal is then combined with the original signal obtained during the decomposition step. Finally, the spatial bands are collapsed to generate

the magnified output video. This method has been successfully applied in various domains, such as visualizing blood flow in facial regions, where pixel color variations and spatial positions are used to identify and enhance motions within specific frequency bands of interest.

5.2.1. Space-time video processing

Space-time video processing is a technique that focuses on magnifying local color changes in a fixed region of a video, enabling the visualization of subtle temporal variations. This approach comprises two main components: spatial processing and temporal processing, both aimed at revealing small changes over time. The spatial processing step involves pooling multiple pixels to enhance the temporal signal-to-noise ratio. To achieve this, a Laplacian pyramid, inspired by the work of Burt and Adelson [94], is employed. The video frames are spatially low-pass filtered and downsampled to optimize computational efficiency. In the temporal processing stage, each spatial band is individually processed. The time series corresponding to the intensity of each pixel within a frequency band is analyzed. A bandpass filter is applied to select specific frequencies of interest. For instance, to magnify pulse rates, a narrow band between 0.4–4 Hz, corresponding to 24–240 beats per minute, is utilized. The temporal processing is consistent across all spatial layers and pixels within each layer. The selected signal is magnified by a specified factor and added back to the original signal. Finally, the pyramid is collapsed to generate the resulting video [102,103].

5.2.2. The concept

The temporal processing technique employed in this study is based on the concept of first-order Taylor series expansions, which has been widely used in the field of optical flow analysis [104]. If we consider a 1D signal undergoing translational motion, with $I(x, t)$ representing the image intensity at position x and time t , and a displacement function $\delta(t)$ representing the motion. In the case of translational motion, the observed intensities can be expressed as $I(x, t) = f(x + \delta(t))$ for the first function and $I(x, 0) = f(x)$ as the second function. To synthesize the signal for motion magnification with a factor of α , the desired output signal (target) can be formulated as follows:

$$\hat{I}(x, t) = f(x + (1 + \alpha)\delta(t)). \quad (1)$$

Assuming a small displacement $\delta(t)$, we can apply a first-order Taylor series expansion about x to approximate the image,

$$I(x, t) \approx f(x) + \delta(t) \frac{\partial f(x)}{\partial x}. \quad (2)$$

Let $B(x, t)$ represent the output obtained by applying a broadband temporal bandpass filter to $I(x, t)$ at each position x , excluding $f(x)$. We assume that $\delta(t)$ represents the motion signal within the temporal bandpass filter,

$$B(x, t) = \delta(t) \frac{\partial f(x)}{\partial x}. \quad (3)$$

Finally, the bandpass signal needs to be amplified by a factor of α and added back to the original signal $I(x, t)$. This results in the magnified signal for this process,

$$\tilde{I}(x, t) = f(x) + (1 + \alpha)\delta(t) \frac{\partial f(x)}{\partial x}. \quad (4)$$

Since $(1 + \alpha)\delta$ is small, a first-order Taylor expansion is still applicable for (4). This yields the following equation:

$$\tilde{I}(x, t) \approx f(x + (1 + \alpha)\delta(t)). \quad (5)$$

This equation represents the magnification of the spatial displacement $\delta(t)$ between the video frames at two time instants t and $t + 1$ (or alternatively, 0 and t) by a factor of $(1 + \alpha)$

5.2.3. Filter selection

The choice of a temporal bandpass filter is crucial in the process of motion magnification. The filter is selected based on the specific application and the desired signals or motions to be magnified within a specific frequency band. For motion magnification, a broad passband is generally recommended to capture a wide range of motions. On the other hand, for color magnification, a narrow passband is often preferred to focus on specific color variations.

5.2.4. Limitations and enhanced approach

The Linear motion magnification approaches, relying on first-order Taylor Expansion, have certain limitations when dealing with high magnification factors or large input motions. These limitations can lead to breakdowns in the magnification process and amplification of noise in the input signal. However, to address these limitations, Liu et al. [105] introduced an enhanced technique that improves the effectiveness of Eulerian video magnification. This state-of-the-art motion magnification approach utilizes spatio-temporal filtering to manipulate small motions in videos. The proposed technique employs the Eulerian video magnification as a video spatio-temporal motion analyzer to generate pixel-level motion mapping. Subsequently, the input video pixels are warped using this mapping to magnify motion without modifying the pixel values. The proposed method [105] supports larger amplification and is substantially less impacted by the frame noise, providing a significant advantage over previous approaches.

5.2.5. Video acceleration magnification

Zhang et al. [30] proposed an alternative approach to address the limitation of large motion in video magnification by utilizing a temporal processing technique analogous to the second-order derivative, known as acceleration [30]. Instead of magnifying all temporal changes, they focus on magnifying the deviation of change. To achieve this, they apply a second-order Taylor series expansion about x to decompose the input signal. Assuming the presence of a linear change component $\delta(t)$ and a non-linear second-order component φ , the input signal can be expressed as $I(x, t) = f(x) + \delta(t) + \varphi(t)$,

$$I(x, t) \approx f(x) + (\delta(t) + \varphi(t)) \frac{\partial f(x)}{\partial x} + (\delta(t) + \varphi(t))^2 \frac{1}{2} \frac{\partial^2 f(x)}{\partial x^2}. \quad (6)$$

Let say $\varphi = 0$ in linear part, while $\delta(t) = 0$ in second-order part.

$$I(x, t) \approx f(x) + \delta(t) \frac{\partial f(x)}{\partial x} + \varphi(t)^2 \frac{1}{2} \frac{\partial^2 f(x)}{\partial x^2}. \quad (7)$$

After applying temporal filtering to extract second-order part (S) and magnifying by α ,

$$\hat{I}(x, t) = I(x, t) + \alpha S(x, t). \quad (8)$$

Where the second-order term is only magnified by a factor α and they use a Laplacian filter to apply a second-order derivative for the video.

5.2.6. Applications and use cases of linear Eulerian video magnification

Linear Eulerian Video Magnification has been used in various applications, including healthcare monitoring, animal physiology, flame morphology detection, and breathing waveform extraction. For example, in one study [36], the authors utilized color magnification to amplify subtle color changes caused by blood pulse on two spots, which were then compared with the results obtained by pulse sensors. This non-invasive method showed strong potential for healthcare monitoring. In another study [11], the authors explored the use of video magnification to extract physiological signals from non-contact videographic material of animals, which are often hidden in subtle changes in color or motion.

In [37], the authors utilized Eulerian Video Magnification to detect changes in the morphology of the flame, allowing for visualization of variations on the flame tip and detection of Smoke Point Height (SPH). In [38], the authors utilized Eulerian Motion Magnification and principal component analysis (PCA) to extract breathing waveforms from upper body movements. This method allows for quick and accurate extraction of various breathing patterns from video without requiring motion tracking, and it can also be used to extract vibration patterns from other objects.

5.3. Phase-based magnification

Wadhwa et al. [2] proposed a Phase-based Eulerian approach to overcome the limitations of the linear approach. This technique offers better representation to handle motions and is less susceptible to noise. The analysis is applied to the phase signal over time at various spatial orientations and scales. However, this approach is computationally intensive and slower compared to other methods.

To achieve this as depicted in Fig. 4, the input video is first decomposed into multi-scales using complex steerable pyramids. The input is then decomposed into a set of complex wavelets corresponding to various positions, scales, and orientations, where each wavelet has a local amplitude and a local phase. A temporal band-pass filter is implemented for each sub-band image, and the filtered phase is magnified by a factor and added back to the authentic phase. Finally, the resulting video is reconstructed, where the phase is magnified, but the magnitude is left unchanged.

5.3.1. The concept

If a one-dimensional (1D) image intensity undergoes translation over time, with displacement denoted by $f(x + \delta(t))$, the modified motion can be represented by $f(x + (1 + \alpha)\delta(t))$. This function can be decomposed into a Fourier series,

$$f(x) = \sum_{w=-\infty}^{\infty} A_w e^{iwx}. \quad (9)$$

After taking into account the translation of image intensity over time,

$$f(x + \delta(t)) = \sum_{w=-\infty}^{\infty} A_w e^{iwx + i\delta(t)w}. \quad (10)$$

By applying a temporal bandpass filter to select the phase component $w(x + \delta(t))$, and removing the DC component wx ,

$$B_w(x, t) = w\delta(t). \quad (11)$$

Subsequently, the bandpassed phase $B_w(x, t)$ is multiplied by a factor α to enhance the phase for the sub-band $S_w(x, t)$, resulting in the motion-amplified sub-band,

$$\hat{S}_w(x, t) := S_w(x, t) e^{i\alpha B_w} = A_w e^{iwx + i\alpha(1 + \alpha)\delta(t)w}. \quad (12)$$

Where $\hat{S}_w(x, t)$ represents a complex sinusoid with a motion factor of $(1 + \alpha)$. After collapsing the pyramids by summing all the sub-bands, the resulting sequence is the amplified sequence $f(x + (1 + \alpha)\delta(t))$.

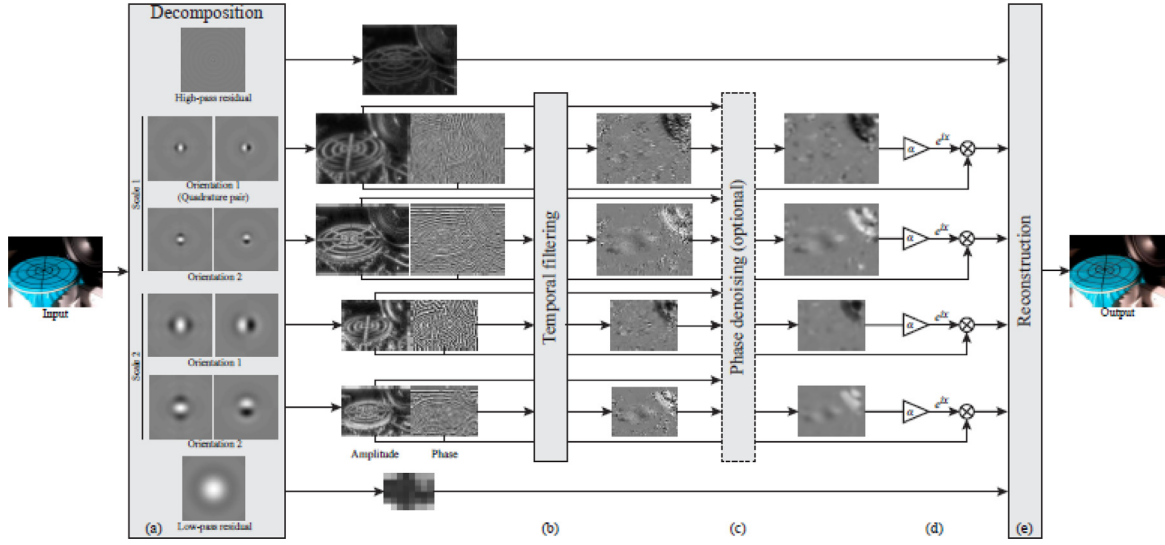


Fig. 4. Infrastructure of the phase video magnification [2].

The complex steerable pyramids effectively capture local motions in a video by calculating the local phase across time for each spatial scale and orientation within a steerable pyramid framework. The phases are then temporally bandpassed to filter out the desired frequency bands relevant to our specific application while removing any temporal DC component. These bandpass phases correspond to motions at different spatial scales and orientations. The synthesized amplified motion is obtained by multiplying these bandpass phases by a magnification factor, resulting in the desired output.

5.3.2. Noise mitigation

As the amplification factor increases, the noise in the signal undergoes a shift rather than being amplified. In the context of the steerable pyramid, a subband is represented as $S_w(x, t)$, and when considering a noisy image $I + \sigma_n n$, the response can be expressed as follows:

$$S_w = e^{i\omega(x+\delta(t))} + \sigma_n N_w(x, t). \quad (13)$$

Let $N_w(x, t)$ represent the response of the noise component n to the complex steerable pyramid filter with index w . Assuming that the magnitude of the noise, σ_n , is much lower than that of the noiseless signal, the temporal filtering for the phase can be expressed as $\omega\delta(t)$. Consequently, the signal after amplification can be described as follows:

$$\hat{S}_w = e^{i\omega(x+(1+\alpha)\delta(t))} + \sigma_n e^{i\alpha\omega\delta(t)} N_w(x, t). \quad (14)$$

The response is shifted by $e^{i\alpha\omega\delta(t)}$. As a result, the noise component experiences a phase shift, indicating a translation in the noise pattern. In contrast to the linear approach [1], the magnitude of the noise remains unchanged, emphasizing that only the phase is affected by the amplification process.

5.3.3. Other phase-based magnification methods

Wadhwa et al. [3] propose an alternative Phase-Based Magnification approach that employs compact and efficient filters in the spatial domain, resulting in a lower overcomplete representation compared to the complex steerable pyramid introduced in [2]. This reduction in overcompleteness enables faster processing, making it suitable for real-time applications. The Riesz transform, a steerable Hilbert transform, is utilized to calculate a quadrature pair that is 90 degrees out of phase with the dominant orientation for each pixel. This phase-shift and translation of features along the dominant orientation of each pixel provide a more focused representation compared to sampling multiple orientations or considering all orientations as in the case of the complex steerable pyramid.

Furthermore, Zhang et al. [30] present another Phase-Based approach that incorporates acceleration magnification for the phase term. In the study by Takeda et al. [40], they introduce a jerk-aware video acceleration magnification technique, influenced by the work of Zhang et al. [30], to amplify subtle changes in both slow and rapid movements. This method utilizes jerk, which corresponds to the third temporal derivative of displacement and signifies the rate of acceleration change over time. In [41], the authors employ a fast local Laplacian filter [42] to improve Eulerian motion magnification by enhancing pyramid decomposition with a focus on edges and fine details, however, this method is not applicable to color magnification. In a different method, Wu et al. [43] introduced Principal Component Analysis (PCA) as a preprocessing step in video magnification. This approach amplifies specific subtle motions in video sequences. Nonetheless, its effectiveness relies on having the significant subtle motions in the input videos more pronounced than irrelevant changes within the principal component. In their paper [44], the authors introduce a video magnification technique specifically designed to detect and amplify significant subtle color or motion variations, even when there is noise in the images. They employ Fractional Anisotropy (FA) to create a novel filter known as the fractional anisotropic filter. This filter selectively enhances meaningful subtle changes while disregarding irrelevant ones. They apply this fractional anisotropic filter to an advanced jerk-aware method [40], resulting in remarkable color or motion magnification outcomes across diverse input video scenarios.

5.3.4. Applications and use cases of phase-based Eulerian video magnification

Phase-Based Eulerian Video Magnification (PB-EVM) has been applied to various fields to enhance visual information and detect subtle movements that are otherwise invisible to the naked eye. In [45], the authors propose a method to extract displacement information from video data by employing motion magnification and optical flow techniques. The study focuses on the challenges associated with interpreting video data in a format familiar to engineers, such as displacement, and demonstrates the effectiveness of the methodology on simple structures. The authors also compare the results with measurements from laser vibrometers and accelerometers, providing further validation of the method.

In this paper [49] the researchers introduce a novel methodology for video camera-based vibration measurement, specifically applied to civil infrastructure assessment. While visual testing has traditionally been a fundamental technique in nondestructive testing (NDT), recent advancements in computer vision and motion magnification [2,3] have

enabled precise quantitative measurements of structural vibrations using video data. Leveraging motion magnification techniques and the capabilities of video cameras, which provide extensive data collection and long-range measurements at the pixel level, this research focuses on improving measurement accuracy. Key areas of emphasis include motion compensation and the effects of video downsampling. The study not only facilitates the detection of smaller subpixel displacements compared to previous camera-based methods but also explores its potential integration into structural health monitoring (SHM) systems for both existing and new structures. The methodology is inspired by the Phase-Based motion magnification technique [2] and is further elaborated in the works of Chen et al. [45].

Chen et al. [50] draw inspiration from motion magnification techniques [45,49] to develop methods for remotely measuring displacements using video cameras. This innovation has the potential to streamline instrumentation processes, making structural health monitoring (SHM) more accessible for various structures. The paper showcases the remote camera-based measurement of motions for the World War I Memorial Bridge in Portsmouth, New Hampshire, a vertical-lift bridge, from a distance exceeding 80 m.

In [51], the authors explore the estimation of material properties from video data, linking vibration mechanics and computer vision. Vibrating objects have distinctive modes of vibration, and their frequencies are influenced by material properties. By analyzing these frequencies in video footage, this study can infer material properties. The approach is demonstrated across various objects using both high-speed and standard frame-rate videos, showcasing its potential in diverse applications, including computer vision, robotics, and structural engineering.

Another application of PB-EVM is in recognizing micro-expressions, as discussed in [46]. The study proposes using motion magnification to exaggerate subtle facial movements and improve recognition rates. Experimental results demonstrate the effectiveness of this approach, providing insights into the upper bounds of effective magnification factors. In the field of emergency situations, where identifying vital signs without direct contact with the victim is crucial, [47] proposes the use of PB-EVM to detect small breathing movements in human bodies. The study compares the effectiveness of two video magnification techniques, intensity-based and phase-based, and analyzes factors such as camera stability, distance, light conditions, and computing time. Furthermore, [48] proposes a novel magnification method that aims to decrease the processing time and remove noise in video magnification techniques.

In [12], the authors introduce a real-time video analysis tool that enables the visualization and magnification of sub-micron rhythmic movements performed by biological cells and their surroundings. The tool was used to explore the periodic motions of non-muscle cells, which are difficult to analyze using conventional microscopy techniques. Results show that fibroblast cells exhibit small fluctuations with a dominant frequency that is dependent on the stiffness of their surrounding substrate. Overall, these studies demonstrate the versatility and potential of PB-EVM in various applications and provide insights into the challenges and future directions in the field. Civera et al. [52] employed phase-based video magnification to detect damage and monitor the health of structures.

5.3.5. Limitations of phase-based (hand-crafted) methods

While phase-based methods, often referred to as hand-crafted methods, have proven effective in motion magnification, they are not without limitations:

- **Sensitivity to Noise:** Phase-based methods are often sensitive to noise and artifacts in the input data. This sensitivity can lead to undesirable amplification of noise, affecting the quality of the motion-magnified output, especially when dealing with low-quality videos.

- **Complexity and Parameter Tuning:** The performance of phase-based methods can be highly dependent on the choice of parameters and filter designs. Fine-tuning these parameters can be a time-consuming and intricate process.
- **Difficulty in Handling Large Motions:** These methods may struggle to effectively magnify very large or rapid motions due to their underlying principles.
- **Limited Application Range:** Phase-based methods may struggle with certain types of motions, such as those with large displacements or complex deformations, limiting their applicability in diverse scenarios.
- **Time-Consuming Processing:** The computational complexity of phase-based techniques can be high, resulting in time-consuming processing, especially when applied to large videos. Consequently, these methods may not be suitable for real-time applications, as observed in previous studies like [1].
- **Ineffectiveness for Real-Time Applications:** Due to their computational demands, phase-based methods may not be suitable for real-time applications that require fast processing.
- **Complex Implementation:** Implementing phase-based methods can be complex and may involve intricate mathematical operations, making them less accessible to those with limited mathematical backgrounds.
- **Dependence on High-Quality Data:** To achieve meaningful results, these methods often rely on high-quality input data, limiting their effectiveness with lower-quality or noisy video sources.

Understanding the limitations of phase-based methods is crucial when considering their applicability, as these limitations can restrict their use in certain scenarios. However, it is noteworthy that significant efforts have been made to address and mitigate these constraints. For instance, Wadhwa et al. [3] leveraged the Riesz pyramid to enhance the performance of their method, increasing its suitability for various applications. Furthermore, Zhang et al. [30] tackled the challenge of accommodating large motions, further expanding the potential use cases of phase-based methods. These research endeavors signify progress in improving the versatility of phase-based methods across a broader range of applications.

5.4. Learning-based motion magnification

Previous approaches in video motion magnification rely on hand-designed filters and derivative filters, such as the Laplacian filter and complex steerable pyramid, to decompose the input video into representations [1,2]. However, Oh et al. [4] introduce a novel approach that leverages neural network technology to directly learn decomposition filters. They train their model using a carefully designed synthetic dataset and a two-frame input. In the training phase, the model learns to amplify the difference (velocity) between successive frames, without the need for temporal filters to separate motions as in previous approaches [1–3].

The learned filters achieve comparable or even superior results with reduced artifacts and effective noise reduction, without relying on temporal filters. Nevertheless, temporal filters can still be employed to further enhance the magnification process and select specific frequency bands of interest. The learned representations from the model can be combined with a linear temporal filter to achieve reasonable magnification results and target motions based on desired frequencies. Moreover, in the test mode, various magnification factors can be applied to the learned representation without the need for retraining the model.

5.4.1. The concept

The following is a concept for the motion magnification approach,

$$\tilde{I}(x, t) = f(x + (1 + \alpha)\delta(x, t)). \quad (15)$$

Where, α is the magnification factor; $\delta(x, t)$ is the motion item to be extracted and manipulated.

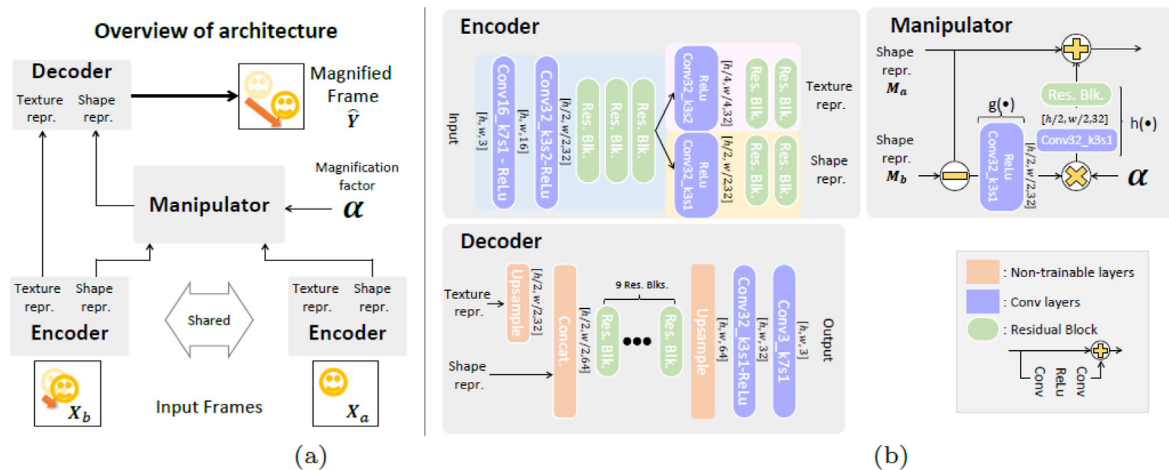


Fig. 5. Deep Convolutional Neural Network Architecture. (a) Architecture Overview. (b) The model has 3 main sections: the encoder, the manipulator, and the decoder. They use two-frame input in training phase [4].

5.4.2. Network architecture

The neural network comprises three fundamental components: the encoder $G_e(\cdot)$, the manipulator $G_m(\cdot)$, and the decoder $G_d(\cdot)$, as depicted in Fig. 5, inspired by previous works [1–3]. The encoder functions as a spatial decomposition filter, extracting a shape representation from a single frame, a process analogous to phase extraction in the steerable and Riesz pyramids [2,3]. The manipulator takes this representation and adjusts it to amplify motion, achieved through a multiplication operation on the difference. Subsequently, the decoder reconstructs the modified representation into the resulting frames with amplified motion. During training, the network takes two video frames, (X_a, X_b) , along with a magnification factor α , as inputs, and produces the magnified frame $\hat{Y}$ as the output. The encoder and decoder modules in the network architecture adopt a fully convolutional design, and residual blocks are incorporated to enhance the overall quality [106]. To ensure a clear separation between texture and shape representation outputs, a regularization loss is employed. Downsampling is achieved through strided convolutions in the encoder to conserve memory, while upsampling is applied in the decoder. In the encoder, a total of three residual blocks are utilized, while the decoder employs nine residual blocks. The texture representation is extracted within the encoder and serves as the secondary output, analogous to the amplitude component in the steerable pyramid [2]. To prevent the amplification of intensity changes, the encoder produces two separate outputs: shape and texture representations. In the decoder, these representations are concatenated again after the manipulation process. The encoder produces two distinct outputs, namely texture represented by $V = G_{e,texture}(X)$ and shape represented by $M = G_{e,shape}(X)$. The manipulator works by taking the difference between shape representations M of two given frames X_a and X_b and directly multiplying a magnification factor α to it, where in the following equation, M is linear with respect to $\delta(x, t)$:

$$G_m(M_a, M_b, \alpha) = M_a + \alpha(M_b - M_a). \quad (16)$$

In practice, some linearity is introduced into the output to improve its quality.

$$G_m(M_a, M_b, \alpha) = M_a + h(\alpha.g(M_b - M_a)). \quad (17)$$

• The function $g(\cdot)$ is implemented as a 3×3 convolution followed by ReLU activation, while $h(\cdot)$ involves a 3×3 convolution followed by a 3×3 residual block.

5.4.3. Loss function

Oh et al. [4] employ the L_1 -loss between the network output $\hat{Y}$ and the actual magnified frame Y . Despite the availability of more sophisticated loss functions like perceptual and adversarial losses [107,

[108], Oh et al. [4] determined that these alternative loss functions did not produce substantial enhancements in their experiments. To emphasize the differentiation between texture and shape representations, intensity perturbations were introduced in specific frames, resulting in the creation of perturbed frames X'_b and Y'_b , where the prime symbol indicates color perturbation. Subsequently, loss functions were applied to compare V'_b with V'_Y (perturbed frames), V_a with V_b (unperturbed frames), and M'_b with M_b , representing the shape representations of the original frames that should remain unchanged. The chosen regularization method was L_1 -loss. Consequently, the network G underwent training by minimizing the comprehensive loss function:

$$L_1(Y, \hat{Y}) + \lambda(L_1(V_a, V_b) + L_1(V'_b, V'_Y) + L_1(M_b, M'_b)), \quad (18)$$

where λ serves as the regularization weight (0.1). This weight parameter, λ , is introduced to control the influence of the regularization term. The optimization process is carried out using the Adam optimizer [109] to minimize the loss function. It is important to note that the original paper contains certain ambiguities and points that are not clearly explained concerning the loss function. These issues will be further discussed in the limitations section to provide clarity and elaboration.

5.4.4. Modes of the model

The network operates in two distinct modes, each with a different frame of reference. During training, the network requires a two-frame input. The first mode, known as the static model, follows the conventional approach of considering the first frame as the reference point. In this mode, the motion is magnified relative to the fixed reference frame. On the other hand, the second mode called the dynamic model, takes into account prior frames as the reference. In this mode, the network amplifies the velocity, which represents the difference between successive frames. Consequently, the magnification factor carries a different meaning in the static and dynamic modes. In the static mode, it signifies the amplification of motion relative to a fixed reference, whereas, in the dynamic mode, it represents the amplification of motion relative to the velocity. During the testing phase, an additional temporal filter is applied to the output.

5.4.5. Temporal filter

The network is initially trained using a two-frame input without the inclusion of a temporal filter. However, the shape representation obtained from the network is amenable to linear temporal filters. In both the Static and Dynamic modes, the difference operation in the manipulator can be substituted with a pixel-wise temporal filter $T_F(\cdot)$ in the manipulator $G_{w(\cdot)}$, where $M(t)$ represents the shape representation

of a video, extracted frame by frame. The manipulator's temporal filter operates as follows:

$$G_{m,temporal}(M(t), \alpha) = M(t) + \alpha T_F(M(t)). \quad (19)$$

The network's decoders combine the texture representation with the output of the manipulator's shape representation after applying temporal filtering. This process results in the generation of output frames with motion that has undergone temporal filtering. This design allows for the selection of the desired frequency band based on the specific requirements of the application.

5.4.6. Synthetic training dataset

A synthetic training dataset was created by Oh et al. [4] to address the challenges of obtaining real motion magnified video pairs for training purposes. The synthetic data was meticulously designed to facilitate effective generalization of the network. The dataset consists of images containing both foreground objects and background scenes. To construct the dataset, a total of 200,000 background images were collected from the MS COCO dataset [110]. Additionally, 7,000 segmented objects were extracted from the PASCAL VOC dataset [111] to serve as foreground objects. These foreground objects were then placed within the background scenes to simulate occlusion effects. This approach helps to preserve sharp object boundaries, although it may potentially result in weaker generalization performance for textures with lower contrast.

To further diversify the dataset, two additional sample types were included. Blurred backgrounds and moving backgrounds were incorporated to simulate scenarios involving large motion. Static scenes were also included to train the network to distinguish between small motion and noise. This was achieved by including complete static scenes as well as static backgrounds with moving foreground objects. To ensure proper learning, certain limitations were imposed on the magnification factor and input motion. The magnification factor was capped at 100, while the maximum allowed input motion was set to 10 pixels to help the network to learn effectively. During the dataset creation process, bicubic interpolation was selected for reconstruction, and the dataset was initially created at a higher resolution to minimize errors introduced during translation using small amounts of motion. Downsampling was then applied to each frame to achieve the desired size, and a Gaussian filter was used to mitigate aliasing artifacts during the downsampling process.

5.4.7. Optimizer

The Adam optimizer [109] is utilized in the training process. It is a stochastic gradient descent algorithm that updates the network weights based on the training data, using adaptive estimation of lower-order moments. The recommended values for the decay rates, beta1, and beta2 are 0.9 and 0.999 respectively, as employed in [4] and our experiments. The learning rate is set to 0.0001, and weight decay is not applied. A batch size of 4 is used, although similar results are obtained with a learning rate of 0.0002 or batch sizes ranging from 4 to 8. To enhance robustness to noise, Poisson noise with random strengths ranging from 0 to 255 pixels is added to the training data.

5.4.8. Magnification factor

The magnification factor, which ranges from 0 to 1, is chosen according to the specific application requirements. It can be adjusted without the need for retraining the model. This flexibility allows for easy customization and adaptation of the magnification level to suit different scenarios.

5.4.9. Limitations

- **Challenges of Temporal Filters in Motion Magnification:** Introducing a temporal filter can enhance output quality by mitigating artifacts, but it also introduces complexities in filter design to ensure compatibility with specific applications and network demands. Furthermore, when the model operates with a temporal filter under conditions of minute motion and high magnification, its performance diminishes, struggling to effectively amplify motion, resulting in artifacts and reduced output quality [4]. The model's original training did not account for the presence of a temporal filter, hence the potential incompatibility. In addition, the model's performance in amplifying slight motions is inferior to hand-designed filters, as illustrated in Figs. 13 and 14. As the magnification factor increases, the challenge of preventing undesired motion amplification and maintaining high-quality output intensifies, particularly when a temporal filter is absent [4].
- **Challenges in Acquiring Sufficient Data Resources:** Learning-based methods have a high demand for extensive and diverse datasets for training. Obtaining and annotating such datasets can be a daunting and expensive task, especially when working with specialized or limited data resources.
- **Challenges in Synthetic Training Datasets:** The development of synthetic training datasets is a painstaking and labor-intensive process essential for effective network training. This complexity and time demand are significant drawbacks to the method. In contrast, conventional approaches [1–3], relying on hand-designed filters, do not necessitate these datasets. Additionally, errors in dataset design and selection can introduce challenges in training, resulting in undesirable artifacts and noise in the output.
- **Dedicated Requirements for Color Magnification: Separate Dataset and Model:** The model [112] and dataset created for motion magnification do not inherently support color magnification. Consequently, a distinct training dataset and model must be developed explicitly to address color magnification tasks. This imposes an additional layer of complexity when expanding the method to include color enhancements.
- **Training Challenges and Computational Demands:** Training deep learning models in motion magnification presents challenges. The process can be computationally demanding, with the official model [112] requiring approximately 20 h on standard hardware, and any parameter modifications necessitating retraining. These constraints limit accessibility on low-end hardware. Additionally, difficulties in hyperparameter tuning, model selection, and achieving convergence can affect the training process.
- **Real-Time Suitability and Testing Time:** The official model by Oh et al. [112] exhibits a runtime of about 20 s for small videos. However, the runtime significantly increases for longer videos, particularly when a temporal filter is employed. This extended processing time limits its suitability for real-time applications compared to alternative methods, such as [3].
- **Overfitting:** Overfitting occurs when a model is excessively tuned to the training data, resulting in poor generalization to new, unseen data. Learning-based motion magnification methods are prone to overfitting, and their performance may deteriorate when applied to real-world scenarios that differ from the training data.
- **Lack of Interpretability:** Deep learning models often lack interpretability. Understanding why a model makes a particular magnification decision can be challenging. In sensitive applications like healthcare, a lack of interpretability may limit the adoption of these methods.
- **Generalization Issues:** Learning-based methods may struggle to generalize well to video data with significantly different characteristics from their training data. For example, if a model is trained on videos with specific lighting conditions, it might not work well in different lighting scenarios.

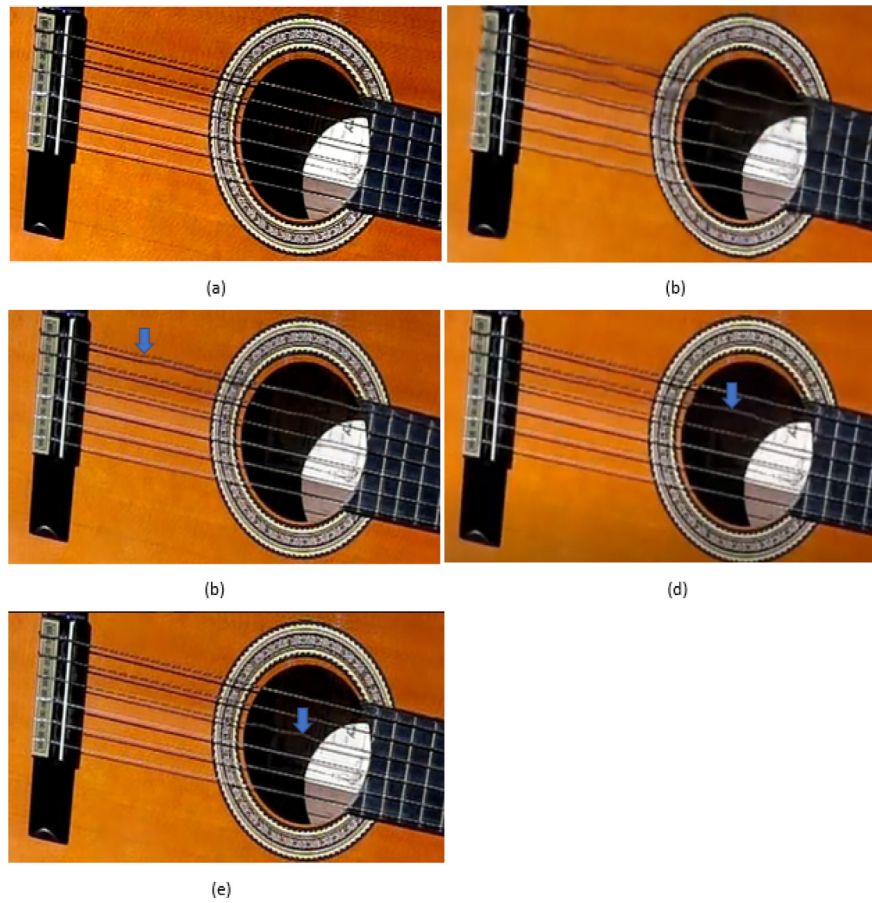


Fig. 6. Quality evaluation of the temporal filter [4]. (a) Input frame. (b) Dynamic mode magnification with a factor of 40, amplifying all motion in the frame. (c, d, e) Temporal filter applied (with a factor of 40) to selectively magnify specific frequency bands for each string.

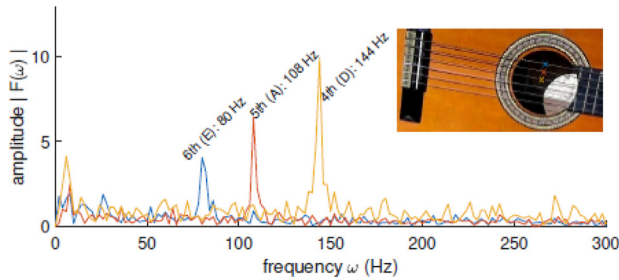


Fig. 7. Intensity signal analysis for each guitar string. Sourced from [4].

- **Error Propagation:** Errors in the initial frames of a video can propagate through the magnification process, leading to significant distortions in the magnified output. Correcting these errors can be challenging and require additional post-processing steps.
- **Inconsistencies in Perturbation and Manipulator Functions:** The discrepancy between the official paper and the dataset is observed in terms of intensity perturbation and the manipulator function. In the paper, intensity perturbation is applied to certain frames for regularization purposes. Frame A (X_a) and frame C (X_b) are unperturbed, while frame B is the perturbed version of frame C (X'_b). However, the meaning of the term V'_Y in the loss function $L_1(V'_b, V'_Y)$ remains unclear as it is not explicitly defined in the code or dataset. Furthermore, the manipulator function differs between the paper and the code. In the paper, it is defined as $G_{m,temporal}(M(t), \alpha) = M(t) + \alpha T_F(M(t))$, whereas in the code, it is

implemented as (*enc_b+diff*) instead of (*enc_a+diff*). The reason behind this change and its impact on the output are uncertain.

Oh et al.'s work [4] offers a promising approach to motion magnification using neural networks. Nonetheless, it is essential to acknowledge certain limitations in this approach. These limitations point to areas that require further research and development for more effective and comprehensive solutions. Recognizing these constraints is pivotal when considering the adoption of learning-based video motion magnification techniques for specific applications. The subsequent sections will delve into ongoing research efforts aimed at enhancing the versatility, efficiency, and reliability of these methods.

5.4.10. Swin transformer based video motion magnification

In this study [31], the authors introduce a new state-of-the-art model called Swin Transformer Based Video Motion Magnification (STB-VMM), inspired by the Learning-based motion magnification VMM (LB-VMM) by Oh et al. [4], but utilizing the Swin Transformer architecture [57]. LB-VMM by Oh et al. introduced a method that replaces hand-designed filters with Convolutional Neural Networks (CNNs) for filter learning but relies on additional temporal filtering to improve image quality. Lado Roig et al.'s [31] approach enhances learnable filters and eliminates temporal filtering, resulting in accurate magnification with improved tolerance to noise, and higher-quality outputs with reduced noise, blurriness, and artifacts compared to previous methods.

The Swin Transformer [57] represents a refined adaptation of the original transformer architecture. Recent developments in computer vision (CV) have successfully employed the Swin transformer for diverse tasks, achieving state-of-the-art results [113,114]. The SwinIR model [114] comprises three key modules: a shallow feature extractor,

a deep feature extractor based on the transformer architecture, and a high-quality image reconstruction module. This architectural design demonstrates exceptional performance across multiple image restoration tasks, including image super-resolution, artifact reduction, and image denoising. These advancements and applications hold particular relevance for motion magnification research.

The STB-VMM model architecture [31] closely resembles that of LB-VMM [4], comprising three primary components: the feature extractor, the manipulator, and the reconstructor. The key distinction lies in the utilization of the Swin Transformer [57]. The effectiveness of the STB-VMM model is assessed by comparing its results with LB-VMM [4] approach across ten distinct video benchmarks as outlined in [31] paper. A sample of this comparison is shown in Fig. 15. However, the new model [31] demands greater computational resources compared to prior models and lacks real-time capabilities like traditional hand-designed filter-based motion magnification methods [1,3].

5.4.11. Other learning-based motion magnification methods and applications

Several studies have leveraged motion magnification and deep learning techniques for some applications such as deepfake detection. In the work by Fei et al. [19], they propose a method based on motion discrepancies to effectively distinguish AI-generated fake videos from genuine ones. This approach employs a two-stage architecture consisting of a CNN+LSTM. Initially, the amplitude of facial motions in videos is magnified, revealing more pronounced distortion or flicker in fake videos compared to authentic ones. They pre-train a deep CNN on frames extracted from training videos, and the output vectors of frame sequences are subsequently utilized as input for an LSTM during the secondary training stage.

In Das et al.'s work [20], they amalgamate traditional Eulerian video magnification [1] with deep learning models, including the Structural Similarity Index (SSIM) and a long short-term memory (LSTM) network. Traditional Eulerian video magnification applies spatial decomposition and temporal filtering to enhance concealed features like skin pulsation and subtle motions in video data. In their approach, features extracted from the Eulerian technique are utilized to train three models for the classification of altered and unaltered videos.

In another study by Ciftci et al. [21], the authors propose a method that leverages the disparity between genuine motion and amplified GAN fingerprints by combining deep learning [4] with traditional motion magnification [2]. This approach aims to detect fake videos and identify their source generators. In authentic videos, the magnified motion aligns with regular human motion patterns, maintaining temporal and spatial correlations between action units and other muscles. Conversely, in fake videos, the generative noise dominates sub-muscular motion. Consequently, when motion is magnified, the generative noise is accentuated rather than the expected human motion patterns.

Furthermore, in the work by Albor et al. [58], a method is introduced that combines traditional approaches with the Hermite Transform and deep learning to monitor heart and respiratory rates. Initially, motion in a video sequence is magnified using the Hermite transform. Concurrently, a deep learning technique is employed to estimate the beat-by-beat pulse signal. The outcomes demonstrate superior enhancement of spectral information from color changes, enabling a more precise estimation of the instantaneous beat-by-beat pulse compared to classical video magnification using a Gaussian pyramid. Also, the study by Lado Roig et al. [6] employs learning-based video motion magnification [4] to automatically extract vibrational signatures, which are used for the identification of structural damage. The papers [59,60] also employ deep learning-based solutions for motion magnification with the aim of reducing artifacts.



Fig. 8. Magnification of a baby video was performed using a magnification factor of 40 in dynamic mode.

6. Results and comparative analysis

In this section, we present the implementation results and comparative analyses of various motion magnification approaches [2–4]. Sections Section 6.1, 6.3, and 6.6 showcase the outcomes of our experiments, conducted using the official code [112] for Oh et al.'s learning-based motion magnification approach [4], with minor configuration adjustments applied to their official code. In contrast, Sections Section 6.2, 6.4, 6.5, and 6.7 present results obtained directly from different approaches outlined in their respective official papers [2–4]. Our analysis primarily emphasizes the learning-based motion magnification approach [4], underlining its innovative nature in comparison to traditional hand-crafted filter methods [2,3].

In our experiments, we have adhered to the recommended parameters specified in [4]. For instance, in cases involving the comparison of two approaches, we have consistently used the same magnification factors as suggested. Regarding the sampling rate, we have followed the guidelines provided in [4]. For example, for baby videos, a sampling rate of 30 fps was employed, while for crane videos, 24 fps was used, and for camera videos, a rate of 300 fps was applied. It is important to note that all the videos utilized in these experiments, as well as the results of said experiments, were sourced from [4], with the exception of the last experiment, for which we utilized data from [31]. The majority of these videos were recorded under optimal conditions, with ample lighting, and using high-definition cameras.

The network proposed by Oh et al. [4,112] is engineered for the purpose of magnifying motion extracted from input videos through the application of a designated magnification factor. The network's architecture is delineated in Fig. 5, and it operates in accordance with the mathematical formulations presented in Eqs. (15) and (17).

6.1. Quality output: Baby video magnification

Figs. 8 and 9 visually demonstrate the exceptional quality of network outputs, showcasing the effectiveness of both non-temporal filtered and temporal filtered results. Nevertheless, it is evident that when a temporal filter is employed, the output quality experiences a notable improvement. The employment of a temporal filter plays a pivotal role in enhancing output quality by notably reducing noise and alleviating unwanted ringing artifacts. This improvement holds particular importance in domains with specific requirements, such as the field of medicine (see Fig. 10).

6.2. Quality comparison: Balance video magnification

The experimental results depicted in Fig. 11 demonstrate the superior handling of occlusion issues by the approach proposed by Oh et al. [4] compared to hand-designed filters [2].



Fig. 9. Baby video magnified with a factor of 40 and utilizing a temporal filter.

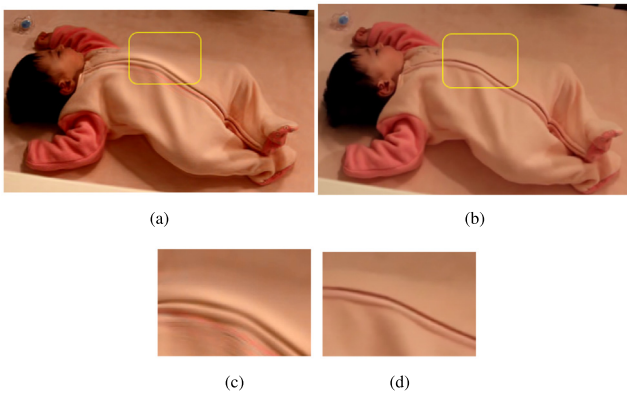


Fig. 10. Quality comparison: (a) and (c) show results of the phase-based approach [2], while (b) and (d) demonstrate the Learning-based Motion Magnification approach [4] with reduced ringing artifacts and enhanced edge details.

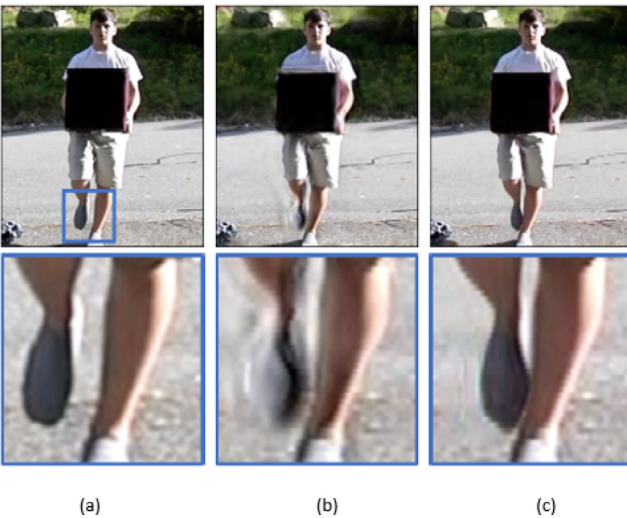


Fig. 11. A quality Comparison. Handling occlusion [4]. (a) Original Frame. (b) Phase-based [2]. (c) Learning-based [4]. Derived from [4].

6.3. Temporal filter analysis on guitar video

The guitar video exhibits oscillations at different frequency bands for each string, as shown in 7. All motion and frequencies are magnified when the model is tested using the dynamic mode. However,

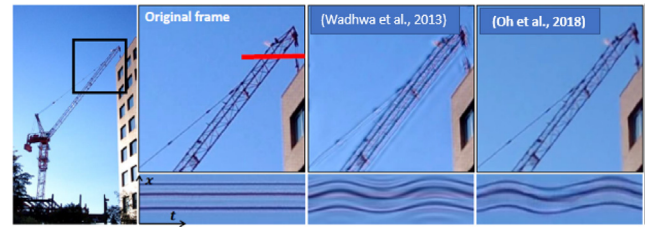


Fig. 12. A quality comparison. A slide appears with a red line over time. Derived from [4].

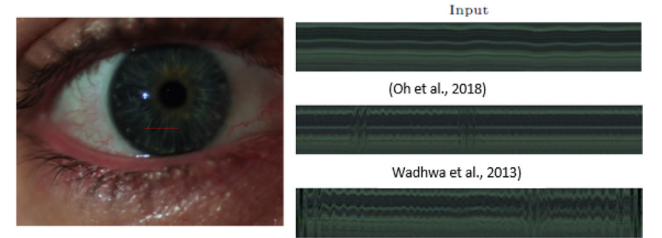


Fig. 13. Quality Comparison: Analysis of a Slide with a Red Line over Time. Sourced from [4].

by employing a temporal filter, it becomes possible to extract specific frequency bands and magnify them selectively for each string, as demonstrated in Fig. 6. The temporal filter used in this analysis is a 2nd-order Butterworth filter [4], which effectively selects and magnifies three specific frequency bands for the guitar strings: (70–90 Hz), (100–120 Hz), and (134–154 Hz). This analysis highlights the effective compatibility of the network [4] with the temporal filter, showcasing the ability to selectively magnify desired frequency bands for each guitar string or motion.

6.4. Magnification of large motion in crane video

The motion in the crane video undergoes magnification with a factor of 75 using the dynamic mode to handle significant motion. Fig. 12 presents an analysis conducted in the original paper [4], depicting a slide with a red line over time. It offers a qualitative comparison between [2,4], demonstrating that [4] produces superior results with reduced noise. This example highlights the potential application of video motion magnification technology in construction scenarios, such as observing subtle movements in swing cranes or suspension bridges affected by wind.

6.5. Challenges with Small Motion: Failures in Eye Video Magnification [4]

There are a few instances where the approach proposed by Oh et al. [4] encounters limitations when using a temporal filter. One such case is observed in an eye video with very subtle motion, which requires the temporal filter to select and magnify this motion. However, the temporal filter employed in [4] fails to amplify this small motion, whereas Wadhwa et al. [2] achieve a successful magnification of the iris motion, as depicted in Fig. 13. This suggests that the compatibility between the trained model and the temporal filter is not entirely optimal. Furthermore, the model occasionally yields unsatisfactory results when dealing with extremely minute motions.

6.6. Challenges with small motion: Failures in camera video magnification

This experiment highlights another failure case observed in a camera video, which involves very subtle motion that requires the use of a temporal filter to extract. However, both the network in [4] operating with or without a temporal filter failed to effectively magnify this

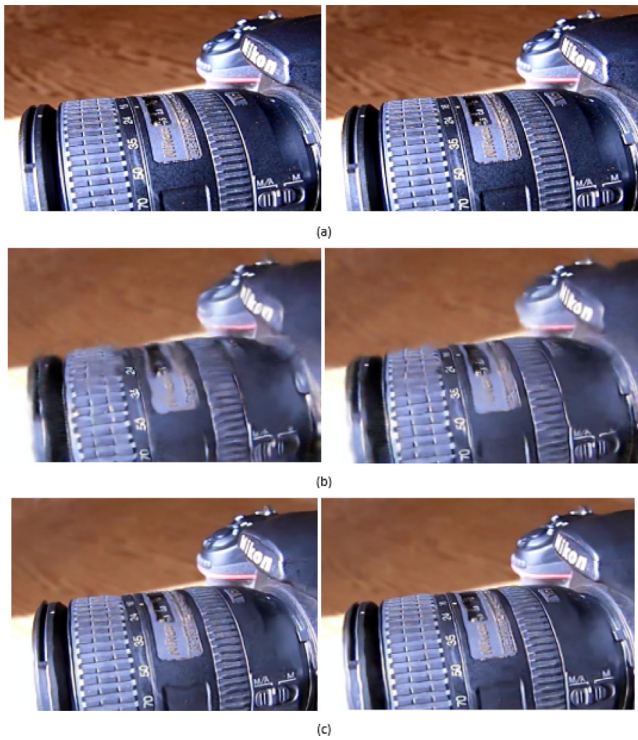


Fig. 14. Failure case of the network with a temporal filter [4]. (a) Original motion. (b) The network (Static model) with a magnification factor of 80 fails to amplify the small motion, resulting in blurred frames. (c) The network with a temporal filter, using a magnification factor of 80, also fails to magnify the small motion.

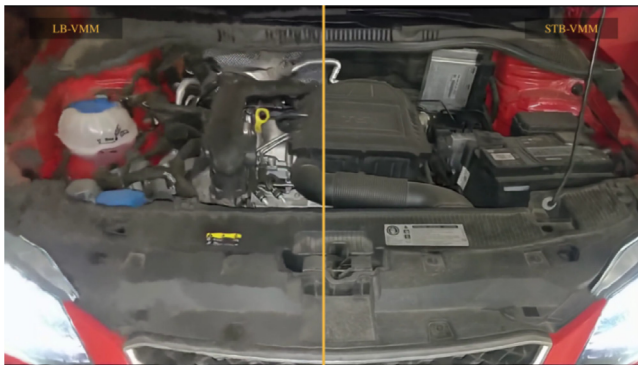


Fig. 15. A quality comparison. LB-VMM (left) [4] vs. STB-VMM (right) [31]. Adapted from [31].

motion, as shown in Fig. 14. The model in [4] exhibits limitations when dealing with very small motions and large magnification factors, especially when a temporal filter is applied. In contrast, the phase-based approach in [2] demonstrates superior results and successfully magnifies the motion.

6.7. Quality comparison: LB-VMM vs. STB-VMM

This experiment provides a comparative analysis of LB-VMM [4] and STB-VMM. Fig. 15 illustrates the comparison, sourced from Lado Roig et al.'s official paper [31]. The figure displays a randomly selected frame from a car video processed by both models. STB-VMM demonstrates superior performance in terms of image clarity, especially regarding edge and texture enhancement, even though the car sequence was recorded in low-light conditions.

7. Conclusions and future work

This paper emphasizes motion magnification techniques, particularly Eulerian approaches, which encompass methods relying on both hand-designed filters like [1–3] and deep neural networks [4,31]. Eulerian approaches are preferred over Lagrangian approaches due to their computational efficiency. In this study, we conducted an analysis of various Eulerian motion magnification approaches, examining their strengths and limitations. Wu et al.'s linear Eulerian approach [1] offers simplicity and speed but struggles with large magnification factors. Phase-based methods, such as those proposed by Wadhwa et al. [2], address these limitations by employing complex steerable pyramids but suffer from slower processing speeds. Riesz pyramids [3] provide an alternative phase-based approach with reduced overcompleteness, enabling real-time application. Elgharib et al. [73] propose a hybrid method to handle large motions, while Zhang et al. [30] utilize acceleration-based temporal processing to address similar challenges. Additional phase-based motion magnification methods [40,42,43] have been introduced to address challenges like noise reduction, real-time applicability, and the handling of significant motions.

Oh et al. [4] introduce a novel approach by replacing hand-designed filters with deep convolutional neural networks, which learn decomposition filters using a two-frame input and a synthetic dataset. However, the performance of the network deteriorates when a temporal filter is applied under high magnification factors and very small motions, leading to limited effectiveness. Several subsequent studies [19,20,31] have been influenced by Oh et al. [4], combining motion magnification techniques with deep learning to leverage the advantages of deep neural networks and eliminate the need for hand-designed filters. However, these approaches may not be well-suited for real-time applications due to their higher computational resource requirements compared to faster MM techniques like those dependent on hand-crafted filters [1,3].

Our analysis and experiments primarily focused on the learning-based approach proposed by Oh et al. [4], which stands out as the first method to leverage the capabilities of neural networks in learning decomposition filters instead of relying on hand-designed filters. While this approach shows promise, however, our findings reveal certain limitations that warrant further research for better understanding and optimization. Addressing these drawbacks will be a key focus of our future research, particularly regarding the compatibility of the trained model with temporal filters and the extension of the model to incorporate color magnification capabilities. Additionally, the development of a comprehensive synthetic dataset will be crucial for training the network in motion magnification applications, considering the difficulty of obtaining real-world datasets for some specific tasks.

CRediT authorship contribution statement

Ahmed Mohamed Ahmed: Contributed to the initial idea, Research methods, Writing, Reviewed and edited the manuscript, Conducted experiments, Developed software, Performed data analysis. **Mohamed Abdelrazek:** Contributed to the initial idea, Reviewed and edited the manuscript, Involved in research methods, Provided supervision, Contributed to the results and discussion sections. **Sunil Aryal:** Reviewed and analyzed the paper, Overall paper discussion, Provided additional supervision. **Thanh Thi Nguyen:** Reviewed and edited the manuscript, particularly in the related works and experiments sections, Provided supervision.

Declaration of competing interest

We declare no conflict of interest in relation to the research presented in this manuscript titled “An Overview of Eulerian Video Motion Magnification Methods”. We appreciate the opportunity to submit this manuscript for review and publication in computers & Graphics Journal.

Data availability

No data was used for the research described in the article.

References

- [1] Wu HY, Rubinstein M, Shih E, Guttaj J, Durand F, Freeman W. Eulerian video magnification for revealing subtle changes in the world. *ACM Trans Graph* 2012;31(4).
- [2] Wadhwa N, Rubinstein M, Durand F, Freeman WT. Phase-based video motion processing. *ACM Trans Graph* 2013;32(4).
- [3] Wadhwa N, Rubinstein M, Durand F, Freeman WT. Riesz pyramids for fast phase-based video magnification. In: *Computational Photography (ICCP), 2014 IEEE International Conference on*. IEEE; 2014.
- [4] Oh TH, Jaroensri R, Kim C, Elgharib M, Durand F, Freeman WT, Matusik W. Learning-based video motion magnification. 2018, arXiv preprint [arXiv:1804.02684](https://arxiv.org/abs/1804.02684).
- [5] Fioriti V, Roselli I, Tati A, Romano R, De Canio G. Motion magnification analysis for structural monitoring of ancient constructions. *Measurement* 2018;129:375–80. [http://dx.doi.org/10.1016/j.measurement.2018.07.055](https://doi.org/10.1016/j.measurement.2018.07.055), URL <https://www.sciencedirect.com/science/article/pii/S0263224118306821>.
- [6] Lado-Roigé R, Font-Moré J, Pérez MA. Learning-based video motion magnification approach for vibration-based damage detection. *Measurement* 2023;206:112218. [http://dx.doi.org/10.1016/j.measurement.2022.112218](https://doi.org/10.1016/j.measurement.2022.112218), URL <https://www.sciencedirect.com/science/article/pii/S0263224122014142>.
- [7] Shao Y, Li L, Li J, An S, Hao H. Target-free 3D tiny structural vibration measurement based on deep learning and motion magnification. *J Sound Vib* 2022;538:117244. [http://dx.doi.org/10.1016/j.jsv.2022.117244](https://doi.org/10.1016/j.jsv.2022.117244), URL <https://www.sciencedirect.com/science/article/pii/S0022460X2200431X>.
- [8] Zona A. Vision-based vibration monitoring of structures and infrastructures: An overview of recent applications. *Infrastructures* 2021;6(1). [http://dx.doi.org/10.3390/infrastructures6010004](https://doi.org/10.3390/infrastructures6010004), URL <https://www.mdpi.com/2412-3811/6/1/4>.
- [9] Balakrishnan G, Durand F, Guttaj J. Detecting pulse from head motions in video. In: *2013 IEEE Conference on Computer Vision and Pattern Recognition*. 2013, p. 3430–7. [http://dx.doi.org/10.1109/CVPR.2013.440](https://doi.org/10.1109/CVPR.2013.440).
- [10] McLeod A, Baxter J, De Ribaupierre S, Peters T. Motion magnification for endoscopic surgery. 9036, 2014, p. 90360C. [http://dx.doi.org/10.1117/12.2043997](https://doi.org/10.1117/12.2043997).
- [11] Lauridsen H, Gonzales S, Hedwig D, Perrin K, Williams C, Wrege P, Bertelsen M, Pedersen M, Butcher J. Extracting physiological information in experimental biology via Eulerian video magnification. *B M C Biology* 2019;17. [http://dx.doi.org/10.1186/s12915-019-0716-7](https://doi.org/10.1186/s12915-019-0716-7).
- [12] Shabi O, Natan S, Kolel A, Mukherjee A, Tchaicheeyan O, Wolfenson H, Kiryati N, Lesman A. Motion magnification analysis of microscopy videos of biological cells. *PLOS ONE* 2020;15(11):1–18, URL [http://dx.doi.org/10.1371/journal.pone.0240127](https://doi.org/10.1371/journal.pone.0240127).
- [13] Janatka M, Marcus HJ, Dorward NL, Stoyanov D. Surgical video motion magnification with suppression of instrument artefacts. 2020, [arXiv:2009.07432](https://arxiv.org/abs/2009.07432).
- [14] Aubakir B, Nurimbetov B, Tursynbek I, Varol HA. Vital sign monitoring utilizing Eulerian video magnification and thermography. In: *2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2016, p. 3527–30. [http://dx.doi.org/10.1109/EMBC.2016.7591489](https://doi.org/10.1109/EMBC.2016.7591489).
- [15] Rossol S, Yang J, Toney-Noland C, Bergin J, Basavaraju C, Kumar P, Lee H. Non-contact video-based neonatal respiratory monitoring. *Children* 2020;7:171. [http://dx.doi.org/10.3390/children7100171](https://doi.org/10.3390/children7100171).
- [16] Oh YH, See J, Le Ngo AC, Phan RCW, Baskaran VM. A survey of automatic facial micro-expression analysis: Databases, methods, and challenges. *Front Psychol* 2018;9:1128. [http://dx.doi.org/10.3389/fpsyg.2018.01128](https://doi.org/10.3389/fpsyg.2018.01128), URL <https://www.frontiersin.org/article/10.3389/fpsyg.2018.01128>.
- [17] Peng W, Hong X, Xu Y, Zhao G. A boost in revealing subtle facial expressions: A consolidated Eulerian framework. In: *2019 14th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2019)*. 2019, p. 1–5. [http://dx.doi.org/10.1109/FG.2019.8756541](https://doi.org/10.1109/FG.2019.8756541).
- [18] Bai M, Goecke R, Herath D. Micro-expression recognition based on video motion magnification and pre-trained neural network. In: *2021 IEEE International Conference on Image Processing (ICIP)*. 2021, p. 549–53. [http://dx.doi.org/10.1109/ICIP42928.2021.9506793](https://doi.org/10.1109/ICIP42928.2021.9506793).
- [19] Fei J, Xia Z, Yu P, Xiao F. Exposing AI-generated videos with motion magnification. *Multimedia Tools Appl* 2020;1–14, URL <https://api.semanticscholar.org/CorpusID:219554813>.
- [20] Das R, Negi G, Smeaton AF. Detecting deepfake videos using Euler video magnification. *Electron Imaging* 2021;33(4):272–1–272-7. [http://dx.doi.org/10.2352/issn.2470-1173.2021.4.mwsf-272](https://doi.org/10.2352/issn.2470-1173.2021.4.mwsf-272).
- [21] Ciftci UA, Demir I. How do deepfakes move? Motion magnification for deepfake source detection. 2022, [arXiv:2212.14033](https://arxiv.org/abs/2212.14033).
- [22] Mehra A, Agarwal A, Vatsa M, Singh R. Motion magnified 3-D residual-indense network for DeepFake detection. *IEEE Trans Biometr, Behav, Identity Sci* 2023;5(1):39–52. [http://dx.doi.org/10.1109/TBIOM.2022.3201887](https://doi.org/10.1109/TBIOM.2022.3201887).
- [23] Selvaraju V, Spicher N, Wang J, Ganapathy N, Warnecke JM, Leonhardt S, Swaminathan R, Deserno T. Continuous monitoring of vital signs using cameras: A systematic review. *Sensors* 2022;22:4097. [http://dx.doi.org/10.3390/s22114097](https://doi.org/10.3390/s22114097).
- [24] Śmieja M, Mamala J, Praznowski K, Ciepliński T, Szumilas T. Motion magnification of vibration image in estimation of technical object condition-review. *Sensors* 2021;21(19). [http://dx.doi.org/10.3390/s21196572](https://doi.org/10.3390/s21196572), URL <https://www.mdpi.com/1424-8220/21/19/6572>.
- [25] Nowara EM, McDuff D, Veeraraghavan A. Combining magnification and measurement for non-contact cardiac monitoring. In: *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. 2021, p. 3805–14. [http://dx.doi.org/10.1109/CVPRW53098.2021.00422](https://doi.org/10.1109/CVPRW53098.2021.00422).
- [26] Le Ngo AC, Phan RC-W. Seeing the Invisible: Survey of Video Motion Magnification and Small Motion Analysis. 52, (6). New York, NY, USA: Association for Computing Machinery; 2019, [http://dx.doi.org/10.1145/3355389](https://doi.org/10.1145/3355389), URL [http://dx.doi.org/10.1145/3355389](https://doi.org/10.1145/3355389).
- [27] Simoncelli E, Freeman W, Adelson E, Heeger D. Shifttable multiscale transforms. *IEEE Trans Inform Theory* 1992;38(2):587–607. [http://dx.doi.org/10.1109/18.119725](https://doi.org/10.1109/18.119725).
- [28] Freeman W, Adelson E. The design and use of steerable filters. *IEEE Trans Pattern Anal Mach Intell* 1991;13(9):891–906. [http://dx.doi.org/10.1109/34.93808](https://doi.org/10.1109/34.93808).
- [29] Shahadi H, Albatat H, Jabbar Al-allaq Z, Thahab A. Eulerian video magnification: a review. *Indones J Electr Eng Comput Sci* 2020;18:799–811. [http://dx.doi.org/10.11591/ijeecs.v18.i2.pp799-811](https://doi.org/10.11591/ijeecs.v18.i2.pp799-811).
- [30] Zhang Y, Pinteal SL, van Gemert JC. Video acceleration magnification. In: *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21–26, 2017*. 2017, p. 502–10.
- [31] Lado-Roigé R, Pérez MA. STB-VMM: Swin transformer based video motion magnification. *Knowl-Based Syst* 2023;269:110493. [http://dx.doi.org/10.1016/j.knsys.2023.110493](https://doi.org/10.1016/j.knsys.2023.110493).
- [32] Liu C, Torralba A, Freeman WT, Durand F, Adelson EH. Motion magnification. *ACM Trans Graph* 2005;24(3):519–26.
- [33] Sun D, Yang X, Liu MY, Kautz J. PWC-net: CNNs for optical flow using pyramid, warping, and cost volume. 2018, [arXiv:1709.02371](https://arxiv.org/abs/1709.02371).
- [34] Kroeger T, Timofte R, Dai D, Gool LV. Fast optical flow using dense inverse search. 2016, CoRR [arXiv:1603.03590](https://arxiv.org/abs/1603.03590), URL [http://arxiv.org/abs/1603.03590](https://arxiv.org/abs/1603.03590).
- [35] Revaud J, Weinzaepfel P, Harchaoui Z, Schmid C. EpicFlow: Edge-preserving interpolation of correspondences for optical flow. CoRR 2015. [arXiv:1501.02565](https://arxiv.org/abs/1501.02565), URL [http://arxiv.org/abs/1501.02565](https://arxiv.org/abs/1501.02565).
- [36] He X, Goubran RA, Liu XP. Using Eulerian video magnification framework to measure pulse transit time. In: *2014 IEEE International Symposium on Medical Measurements and Applications (MeMeA)*. 2014, p. 1–4. [http://dx.doi.org/10.1109/MeMeA.2014.6860029](https://doi.org/10.1109/MeMeA.2014.6860029).
- [37] Pino J, Garcés HO, Cuevas J, Arias LE, Rojas AJ, Fuentes A. Soot propensity by image magnification and artificial intelligence. *Fuel* 2018;225:256–65. [http://dx.doi.org/10.1016/j.fuel.2018.03.141](https://doi.org/10.1016/j.fuel.2018.03.141), URL <https://www.sciencedirect.com/science/article/pii/S0016236118305490>.
- [38] Zhang Y, Shang F. Noncontact extraction of breathing waveform. In: *Proceedings of the 2015 International Power, Electronics and Materials Engineering Conference*. Atlantis Press; 2015/05, p. 782–7. [http://dx.doi.org/10.2991/ipemec-15.2015.145](https://doi.org/10.2991/ipemec-15.2015.145), URL [http://dx.doi.org/10.2991/ipemec-15.2015.145](https://doi.org/10.2991/ipemec-15.2015.145).
- [39] Davis A, Rubinstein M, Wadhwa N, Mysore G, Durand F, Freeman WT. The visual microphone: Passive recovery of sound from video. *ACM Trans Graph (Proc. SIGGRAPH)* 2014;33(4):79:1–79:10.
- [40] Takeda S, Okami K, Mikami D, Isogai M, Kimata H. Jerk-aware video acceleration magnification. In: *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Los Alamitos, CA, USA: IEEE Computer Society; 2018, p. 1769–77. [http://dx.doi.org/10.1109/CVPR.2018.00190](https://doi.org/10.1109/CVPR.2018.00190), URL <https://doi.ieeecomputersociety.org/10.1109/CVPR.2018.00190>.
- [41] Verma M, Raman S. Edge-aware spatial filtering-based motion magnification. In: Chaudhuri BB, Kankanhalli MS, Raman B, editors. *Proceedings of 2nd International Conference on Computer Vision & Image Processing*. Singapore: Springer Singapore; 2018, p. 117–28.
- [42] Paris S, Hasinoff SW, Kautz J. Local Laplacian filters: Edge-aware image processing with a Laplacian pyramid. *Commun ACM* 2015;58(3):81–91. [http://dx.doi.org/10.1145/2723694](https://doi.org/10.1145/2723694), URL [http://dx.doi.org/10.1145/2723694](https://doi.org/10.1145/2723694).
- [43] Wu X, Yang X, Jin J, Yang Z. PCA-based magnification method for revealing small signals in video. *Signal, Image Video Process* 2018;12:1293–9, URL <https://api.semanticscholar.org/CorpusID:207317084>.
- [44] Takeda S, Akagi Y, Okami K, Isogai M, Kimata H. Video magnification in the wild using fractional anisotropy in temporal distribution. In: *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, p. 1614–22. [http://dx.doi.org/10.1109/CVPR.2019.00171](https://doi.org/10.1109/CVPR.2019.00171).
- [45] Chen JG, Wadhwa N, Cha YJ, Durand F, Freeman WT, Buyukozturk O. Modal identification of simple structures with high-speed video using motion magnification. *J Sound Vib* 2015;345:58–71. [http://dx.doi.org/10.1016/j.jsv.2015.01.024](https://doi.org/10.1016/j.jsv.2015.01.024), URL <https://www.sciencedirect.com/science/article/pii/S0022460X1500070X>.

- [46] Le Ngo AC, Oh Y-H, Phan RCW, See J. Eulerian emotion magnification for subtle expression recognition. In: 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). 2016, p. 1243–7. <http://dx.doi.org/10.1109/ICASSP.2016.7471875>.
- [47] Ordóñez C, Cabo C, Menéndez A, Bello A. Detection of human vital signs in hazardous environments by means of video magnification. *PLoS One* 2018;13(4):e0195290. <http://dx.doi.org/10.1371/journal.pone.0195290>, URL <http://dx.doi.org/10.1371/journal.pone.0195290>.
- [48] Al-Naji A, Lee SH, Chahl J. An efficient motion magnification system for real-time applications. *Mach Vis Appl* 2018;29(4):585–600. <http://dx.doi.org/10.1007/s00138-018-0916-0>, URL <http://dx.doi.org/10.1007/s00138-018-0916-0>.
- [49] Chen JG, Davis A, Wadhwa N, Durand F, Freeman WT, Büyükoztürk O. Video camera-based vibration measurement for civil infrastructure applications. *J Infrastruct Syst* 2017;23(3):B4016013.
- [50] Chen JG, Adams TM, Sun H, Bell ES, Büyükoztürk O. Camera-based vibration measurement of the world war I memorial bridge in Portsmouth, New Hampshire. *J Struct Eng* 2018. URL <https://api.semanticscholar.org/CorpusID:117158501>.
- [51] Davis* A, Bouman* KL, Chen JG, Rubinstein M, Büyükoztürk O, Durand F, Freeman WT. Visual vibrometry: Estimating material properties from small motions in video. *IEEE Trans Pattern Anal Mach Intell* 2017;39(4):732–45. <http://dx.doi.org/10.1109/TPAMI.2016.2622271>.
- [52] Civera M, Zanotti Fragonara L, Surace C. An experimental study of the feasibility of phase-based video magnification for damage detection and localisation in operational deflection shapes. *Strain* 2020;56. <http://dx.doi.org/10.1111/str.12336>.
- [53] Takeda S, Niwa K, Isogawa M, Shimizu S, Okami K, Aono Y. Bilateral video magnification filter. In: 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2022, p. 17348–57. <http://dx.doi.org/10.1109/CVPR52688.2022.01685>.
- [54] Tomasi C, Manduchi R. Bilateral filtering for gray and color images. In: Sixth International Conference on Computer Vision (IEEE Cat. No.98CH36271). 1998, p. 839–46. <http://dx.doi.org/10.1109/ICCV.1998.710815>.
- [55] Paris S, Durand F. A fast approximation of the bilateral filter using a signal processing approach. In: Leonardis A, Bischof H, Pinz A, editors. *Computer Vision – ECCV 2006*. Berlin, Heidelberg: Springer Berlin Heidelberg; 2006, p. 568–80.
- [56] Koij JFP, van Gemert JC. Depth-aware motion magnification. In: Leibe B, Matas J, Sebe N, Welling M, editors. *Computer Vision – ECCV 2016*. Cham: Springer International Publishing; 2016, p. 467–82.
- [57] Liu Z, Lin Y, Cao Y, Hu H, Wei Y, Zhang Z, Lin S, Guo B. Swin transformer: Hierarchical vision transformer using shifted windows. *CoRR* 2021. [arXiv: 2103.14030](https://arxiv.org/abs/2103.14030), URL <https://arxiv.org/abs/2103.14030>.
- [58] Moya-Albor E, Brieve J, Ponce H, Rivas-Scott O, Gómez-Peña C. Heart rate estimation using Hermite transform video magnification and deep learning. In: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). 2018, p. 2595–8. <http://dx.doi.org/10.1109/EMBC.2018.8512879>.
- [59] Singh J, Murala S, Kosuru GSR. Lightweight network for video motion magnification. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV). 2023, p. 2041–50.
- [60] Singh J, Murala S, Kosuru GSR. Multi domain learning for motion magnification. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2023, p. 13914–23.
- [61] Eitner M, Miller B, Sirohi J, Tinney C. Effect of broad-band phase-based motion magnification on modal parameter estimation. *Mech Syst Signal Process* 2021;146:106995. <http://dx.doi.org/10.1016/j.ymssp.2020.106995>, URL <https://www.sciencedirect.com/science/article/pii/S0888327020303812>.
- [62] Kwasniewska A, Ruminski J, Szankin M. Improving accuracy of contactless respiratory rate estimation by enhancing thermal sequences with deep neural networks. *Appl Sci* 2019;9(20). <http://dx.doi.org/10.3390/app9204405>, URL <https://www.mdpi.com/2076-3417/9/20/4405>.
- [63] Markwirth A, Lachetta M, Mönkemöller V, Heintzmann R, Hübner W, Huser T, Müller M. Video-rate multi-color structured illumination microscopy with simultaneous real-time reconstruction. *Nat News* 2019. URL <https://www.nature.com/articles/s41467-019-12165-x#citeas>.
- [64] Mattioli V, Alinovi D, Ferrari G, Pisani F, Raheli R. Motion magnification algorithms for video-based breathing monitoring. *Biomed Signal Process Control* 2023;86:105148. <http://dx.doi.org/10.1016/j.bspc.2023.105148>, URL <https://www.sciencedirect.com/science/article/pii/S1746809423005815>.
- [65] Uryga A, Kazimierska A, Poppek M, Dragan B, Burzyńska M, Masalski M, Kasprzowicz M. Applying video motion magnification to reveal spontaneous tympanic membrane displacement as an indirect measure of intracranial pressure in patients with brain pathologies. *Acta Neurochirurgica* 2023;165:1–9. <http://dx.doi.org/10.1007/s00701-023-05681-9>.
- [66] Popek MP, Danielewska ME, Iskander DR. Assessing the feasibility of the use of video motion magnification for measuring microdisplacements. *IEEE Trans Instrum Meas* 2017;66(9):2329–36. <http://dx.doi.org/10.1109/TIM.2017.2700118>.
- [67] Wang J, Drucker SM, Agrawala M, Cohen MF. The cartoon animation filter. In: ACM SIGGRAPH 2006 Papers. SIGGRAPH '06, New York, NY, USA: Association for Computing Machinery; 2006, p. 1169–73. <http://dx.doi.org/10.1145/1179352.1142010>, URL <http://dx.doi.org/10.1145/1179352.1142010>.
- [68] Ngo ACL, Johnston A, Phan RC, See J. Micro-expression motion magnification: Global Lagrangian vs. Local Eulerian approaches. In: 13th IEEE International Conference on Automatic Face & Gesture Recognition, FG 2018, Xi'an, China, May 15–19, 2018. 2018, p. 650–6.
- [69] Flotho P, Bhamborae MJ, Haab L, Strauss DJ. Lagrangian motion magnification revisited: Continuous, magnitude driven motion scaling for psychophysiological experiments. In: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). 2018, p. 3586–9. <http://dx.doi.org/10.1109/EMBC.2018.8512892>.
- [70] Flotho P, Heiss C, Steidl G, Strauss DJ. Lagrangian motion magnification with double sparse optical flow decomposition. 2022, [arXiv:2204.07636](https://arxiv.org/abs/2204.07636).
- [71] Sun S, Tang Y, Shi Y, Guo Y, Tang T, Wu Y. An efficient warp-based motion magnification method to reveal subtle changes in video. In: Proceedings of the 4th International Conference on Advanced Information Science and System. AISS '22, New York, NY, USA: Association for Computing Machinery; 2023, <http://dx.doi.org/10.1145/3573834.3574476>, URL <http://dx.doi.org/10.1145/3573834.3574476>.
- [72] Teed Z, Deng J. RAFT: Recurrent all-pairs field transforms for optical flow. 2020, [arXiv:2003.12039](https://arxiv.org/abs/2003.12039).
- [73] Elgharib M, Hefeeda M, Durand F, Freeman B. Video magnification in presence of large motions. In: The IEEE Conference on Computer Vision and Pattern Recognition (CVPR). 2015, p. 4119–27.
- [74] Shi J, Tomasi C. Good Features to Track. *Tech. Rep.*, USA: Cornell University; 1993.
- [75] Verma M, Raman S. Interest Region Based Motion Magnification. 2017, p. 27–39. http://dx.doi.org/10.1007/978-3-319-68560-1_3.
- [76] Gao S, Feng Y, Yang L, Liu X, Zhu Z, Doermann D, Zhang B. MagFormer: Hybrid video motion magnification transformer from Eulerian and Lagrangian perspectives. In: 33rd British Machine Vision Conference 2022, BMVC 2022, London, UK, November 21–24, 2022. BMVA Press; 2022, URL <https://bmvc2022.mpi-inf.mpg.de/0444.pdf>.
- [77] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser L, Polosukhin I. Attention is all you need. *CoRR* 2017. [arXiv:1706.03762](https://arxiv.org/abs/1706.03762), URL <https://arxiv.org/abs/1706.03762>.
- [78] Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, Uszkoreit J, Houlsby N. An image is worth 16x16 words: Transformers for image recognition at scale. *CoRR* 2020. [arXiv:2010.11929](https://arxiv.org/abs/2010.11929), URL <https://arxiv.org/abs/2010.11929>.
- [79] Chen L, Peng C, Zhao B. Novel multi-task learning for motion magnification. *IEEE Trans Circuits Syst Video Technol* 2023. <http://dx.doi.org/10.1109/TCSVT.2023.3262777>, 1.
- [80] Chen L, Zhao B, Liu X, Cai Z, Peng C. Multi-task learning using optical flow for motion magnification. In: Xiao L, Xue J, editors. Fourteenth International Conference on Graphics and Image Processing (ICGIP 2022). 12705, SPIE, International Society for Optics and Photonics; 2023, p. 127052V. <http://dx.doi.org/10.1117/12.2680446>, URL <http://dx.doi.org/10.1117/12.2680446>.
- [81] Poh MZ, McDuff DJ, Picard RW. Non-contact, automated cardiac pulse measurements using video imaging and blind source separation. *Opt Express* 2010;18(10):10762–74.
- [82] Lee J, Chai J, Reitsma PSA, Hodgins JK, Pollard NS. Interactive control of avatars animated with human motion data. *ACM Trans Graph* 2002;21(3):491–500.
- [83] Pullen K, Bregler C. Motion capture assisted animation: Texturing and synthesis. *ACM Trans Graph* 2002;21(3):501–8.
- [84] Brand M, Hertzmann A. Style machines. In: Proceedings of the 27th Annual Conference on Computer Graphics and Interactive Techniques. SIGGRAPH '00, USA: ACM Press/Addison-Wesley Publishing Co.; 2000, p. 183–92.
- [85] Brostow GJ, Essa I. Image-based motion blur for stop motion animation. In: Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques. SIGGRAPH '01, New York, NY, USA: Association for Computing Machinery; 2001, p. 561–6.
- [86] Schödl A, Szeliski R, Salesin DH, Essa I. Video textures. In: Proceedings of the 27th Annual Conference on Computer Graphics and Interactive Techniques. SIGGRAPH '00, USA: ACM Press/Addison-Wesley Publishing Co.; 2000, p. 489–98.
- [87] Wang JY, Adelson EH. Representing moving images with layers. *Trans Img Proc* 1994;3(5):625–38.
- [88] Jojic N, Frey BJ. Learning flexible sprites in video layers. In: 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2001), with CD-ROM, 8–14 December 2001, Kauai, HI, USA. 2001, p. 199–206.
- [89] Butcher JC. Numerical methods for ordinary differential equations. 2003.
- [90] Cheney EW, Kincaid DR. Numerical Mathematics and Computing. 6th. USA: Brooks/Cole Publishing Co.; 2007.
- [91] Kreyszig E, Kreyszig H, Norminton EJ. Advanced Engineering Mathematics. Tenth. Hoboken, NJ: Wiley; 2011.

- [92] Burt PJ, Adelson EH. The Laplacian pyramid as a compact image code. *IEEE Trans Commun* 1983;31(4):532–40. <http://dx.doi.org/10.1109/TCOM.1983.1095851>.
- [93] Burt PJ, Adelson EH. A multiresolution spline with application to image mosaics. *ACM Trans Graph* 1983;2(4):217–36. <http://dx.doi.org/10.1145/245.247>, URL <http://dx.doi.org/10.1145/245.247>.
- [94] Adelson EH, Anderson CH, Bergen JR, Burt PJ, Ogden JM. Pyramid methods in image processing. *RCA Engineer* 1984;29(6):33–41.
- [95] Mallat SG. A theory for multiresolution signal decomposition: The wavelet representation. 1989;11(7):674–93. <http://dx.doi.org/10.1109/34.192463>, URL <http://dx.doi.org/10.1109/34.192463>.
- [96] Daubechies I. The wavelet transform, time-frequency localization and signal analysis. *IEEE Trans Inform Theory* 1990;36(5):961–1005.
- [97] Simoncelli EP, Freeman WT. Steerable pyramids. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 1995, p. 451–8.
- [98] Wiegand T, Girod B, Stuhlmüller J. Image coding using the steerable pyramid transform. *IEEE Trans Image Process* 2001;10(3):465–72.
- [99] Bovik AC. *The Essential Guide To Image Processing*. Inc.: Academic Press; 2009.
- [100] Badin G, Crisciani F. *Variational Formulation of Fluid and Geophysical Fluid Dynamics: Mechanics, Symmetries and Conservation Laws*. Cham: Springer International Publishing; 2018.
- [101] Batchelor GK. *An Introduction To Fluid Dynamics*. Cambridge University Press; 2000.
- [102] Wu Y. Linear-based Eulerian motion modulation. 2014, URL <https://patents.google.com/patent/US20140072190A1/en>.
- [103] Wadhwa N, Wu H-Y, Davis A, Rubinstein M, Shih E, Mysore GJ, Chen JG, Buyukozturk O, Guttag JV, Freeman WT, Durand F. Eulerian video magnification and analysis. *Commun ACM* 2016;60(1):87–95. <http://dx.doi.org/10.1145/3015573>, URL <http://dx.doi.org/10.1145/3015573>.
- [104] Horn BKP, Schunck BG. Determining optical flow. *Artificial Intelligence* 1981;17(1–3):185–203.
- [105] Liu L, Lu L, Luo J, Zhang J, Chen X. Enhanced Eulerian video magnification. In: *2014 7th International Congress on Image and Signal Processing*. 2014, p. 50–4. <http://dx.doi.org/10.1109/CISP.2014.7003748>.
- [106] Shelhamer E, Long J, Darrell T. Fully convolutional networks for semantic segmentation. *IEEE Trans Pattern Anal Mach Intell* 2017;39(4):640–51.
- [107] Isola P, Zhu J, Zhou T, Efros AA. Image-to-image translation with conditional adversarial networks. 2016, [arXiv:1611.07004](https://arxiv.org/abs/1611.07004), URL <http://arxiv.org/abs/1611.07004>.
- [108] Johnson J, Alahi A, Fei-Fei L. Perceptual losses for real-time style transfer and super-resolution. 2016, [arXiv:1603.08155](https://arxiv.org/abs/1603.08155), URL <http://arxiv.org/abs/1603.08155>.
- [109] Kingma DP, Ba J. Adam: A method for stochastic optimization. *CoRR* 2014. URL <http://arxiv.org/abs/1412.6980>.
- [110] Lin T-Y, Maire M, Belongie S, Bourdev L, Girshick R, Hays J, Perona P, Ramanan D, Zitnick CL, Dollár P. Microsoft COCO: Common objects in context. 2014, URL <http://arxiv.org/abs/1405.0312>.
- [111] Everingham M, Gool L, Williams CK, Winn J, Zisserman A. The pascal visual object classes (VOC) challenge. *Int. J. Comput. Vision* 2010;88(2):303–38.
- [112] Oh TH, Jaroensri R, Kim C, Elgharib M, Durand F, Freeman WT, Matsumik W. Learning-based video motion magnification. 2018, https://github.com/12dmodel/deep_motion_mag.
- [113] Han K, Wang Y, Chen H, Chen X, Guo J, Liu Z, Tang Y, Xiao A, Xu C, Xu Y, Yang Z, Zhang Y, Tao D. A survey on vision transformer. *IEEE Trans Pattern Anal Mach Intell* 2023;45(1):87–110. <http://dx.doi.org/10.1109/TPAMI.2022.3152247>.
- [114] Liang J, Cao J, Sun G, Zhang K, Gool LV, Timofte R. Swinir: Image restoration using swin transformer. 2021, [arXiv:2108.10257](https://arxiv.org/abs/2108.10257).